

Exact Affine Geometry, Central Radii, and Certified Compression Bounds for Finite-Outcome Quantum Instruments

Anonymous

Abstract

We determine several exact geometric parameters of the compact convex body of n -outcome quantum instruments between finite-dimensional quantum systems, together with certified compression bounds for continuous encoding. The affine dimension is

$$D_{A,B}^{(n)} = d_A^2(nd_B^2 - 1).$$

Assume $nd_B > 1$, so that $\text{Inst}_n(A, B)$ is not a singleton; when $n = d_B = 1$ the body is the single trace functional, $D_{A,B}^{(1)} = 0$, its direction space is $\{0\}$, every compression width is zero, and the centred relative in-radius is $+\infty$ under the supremum convention.

For $nd_B > 1$, the centred relative instrument-diamond in-radius about the uniform depolarising instrument $\mathcal{I}_*$ is exactly

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_*) = \frac{2}{nd_B \min\{d_A, d_B\}},$$

and the global covering radius is exactly

$$R_{\text{inst}}(\mathcal{I}_*) = 2 \left(1 - \frac{1}{nd_B \min\{d_A, d_B\}} \right),$$

with $\mathcal{I}_*$ an optimal centre and the diameter of the body exactly 2. No centre of the body carries a larger relative ball: the global in-radius, i.e. the supremum over all centres, equals the centred value, so $\mathcal{I}_*$ is an optimal in-radius centre as well. The replacement face has exact antipodal profile

$$\alpha_r(\text{RepInst}_n(A, B)) = \begin{cases} 1, & 0 \leq r \leq nd_B^2 - 2, \\ 0, & r \geq nd_B^2 - 1. \end{cases}$$

We prove a full-direction antipodal construction with separation $2/d_A$, giving

$$\alpha_r(\text{Inst}_n(A, B)) \geq \frac{1}{d_A} \quad (0 \leq r < D_{A,B}^{(n)}),$$

and observable-sphere and orthogonal-join constructions extending the separation-2 regime to an explicit dimension $L_{A,B}^{(n)}$. These certificates combine into a strengthened certified lower envelope $\beta_r^{\text{best}, (n)}(A, B)$, and, together with certified upper bounds from constant codes, convex projection, and balanced input/output pinching, produce the explicit bracket

$$\beta_r^{\text{best}, (n)}(A, B) \leq \delta_{r, \text{inst}}^{\diamond}(A, B) \leq U_r^{(n)}(A, B).$$

Under the convention of continuous encoding with unrestricted decoding, the zero-error Euclidean threshold is exactly the affine dimension:

$$\delta_{r, \text{inst}}^{\diamond}(A, B) = 0 \iff r \geq D_{A,B}^{(n)},$$

and for $r \geq D_{A,B}^{(n)}$ exact decoding may be chosen continuously via Euclidean nearest-point projection in affine coordinates. Every radius and width carries a single-shot operational interpretation through the equal-prior discrimination formula $p_{\text{succ}} = \frac{1}{2} + \frac{1}{4} \|\cdot\|_{\diamond, \text{inst}}$. The single-outcome case recovers channels, and the one-dimensional-output case recovers finite-outcome measurements; the $1/n$ suppression of the centred radius against the n -fold growth of the affine dimension is made explicit.

Contents

1	Introduction	2
2	Preliminaries	2
3	Exact affine and metric parameters	4
4	Antipodal certificates and the lower envelope	11
5	Compression widths: upper bounds and exact thresholds	18
6	Specialisations, examples, and explicit trade-off	25
A	Finite-program evaluation: background	27

1 Introduction

Quantum instruments are the natural generalisation of quantum channels to finite-outcome statistics: a collection of completely positive, trace-nonincreasing maps whose sum is a channel. They describe both the probability of an outcome and the conditional post-measurement state, and they model finite-outcome measurements and adaptive protocols [3, 4, 5].

The question treated here is the same as for channels: can the compact convex body of n -outcome instruments between A and B be encoded continuously into a low-dimensional latent space and decoded with uniformly small error? Throughout, we adopt the standard asymmetric convention: the *encoder is continuous* and the *decoder is unrestricted* (Convention 5.1); where a continuous decoder is additionally available we say so explicitly. The answer has two regimes. At small latent dimension, input-independent preparation of a flagged output state already forces error one. At larger dimensions, the full input-dependent instrument body imposes a ball bound scaling as $1/n$, alongside a full-direction antipodal obstruction that is independent of n .

The main contribution of this paper is the exact affine geometry of the instrument body and its compression consequences; among the exact parameters, the uniform depolarising instrument is an optimal centre for both the covering radius and the global in-radius. Which quantities are determined *exactly*, and which only by *certified bounds*, is stated once in Remark 2.6 and maintained throughout. A separate physical question — whether an abstract code can be realised by a fixed programmable processor — is addressed in Appendix A: exact finite-dimensional evaluation of all unitary channels is impossible, and vanishing error forces unbounded program dimension. This separates *representational* dimension from *programmable* evaluation; the main theorems do not rely on the appendix.

2 Preliminaries

Throughout, A and B are finite-dimensional Hilbert spaces of dimensions d_A, d_B , and $n \in \mathbb{N}$ is the number of outcomes. Let $A_0 \simeq A$ be a reference copy with a fixed pair of conjugate bases in A_0 and A , and define the maximally entangled state

$$|\omega_A\rangle = \frac{1}{\sqrt{d_A}} \sum_{j=1}^{d_A} |j\rangle_{A_0} |j\rangle_A.$$

For a map $\Phi : L(A) \rightarrow L(B)$ the *normalised Choi operator* is

$$\hat{J}_\Phi = (\text{id}_{A_0} \otimes \Phi)(|\omega_A\rangle\langle\omega_A|),$$

the Choi–Jamiołkowski isomorphism in the normalised form [1, 2].

Remark 2.1 (Normalised Choi convention). *With the unnormalised maximally entangled vector $|\Omega_A\rangle = \sum_j |j\rangle_{A_0} |j\rangle_A$ one has $\widehat{J}_\Phi = \frac{1}{d_A} J_\Phi^{\text{unnorm}}$. Consequently Φ is trace preserving if and only if*

$$\text{Tr}_B \widehat{J}_\Phi = \frac{I_{A_0}}{d_A}.$$

All affine constraints, radii, covering bounds, and pinching dimension counts in this paper use this normalised convention.

Let C_n be an n -dimensional classical flag space with basis $|1\rangle, \dots, |n\rangle$.

Definition 2.2 (Instrument, flagged channel, and instrument diamond norm). *An n -outcome instrument from A to B is a tuple $\mathcal{I} = (\mathcal{I}_1, \dots, \mathcal{I}_n)$ of completely positive, trace-nonincreasing maps whose sum is a channel. Write $\text{Inst}_n(A, B)$ for the set of all such instruments. The flagged channel of $\mathcal{I}$ is*

$$\widehat{\mathcal{I}}(X) = \sum_{k=1}^n \mathcal{I}_k(X) \otimes |k\rangle\langle k|.$$

For a tuple $\vec{\Delta} = (\Delta_1, \dots, \Delta_n)$ of Hermiticity-preserving maps, define the flagged map and the instrument diamond norm by

$$\widehat{\vec{\Delta}}(X) = \sum_{k=1}^n \Delta_k(X) \otimes |k\rangle\langle k|, \quad \|\vec{\Delta}\|_{\diamond, \text{inst}} = \|\widehat{\vec{\Delta}}\|_{\diamond}.$$

By block diagonality this equals the operational formula

$$\|\vec{\Delta}\|_{\diamond, \text{inst}} = \sup_{\rho \in \mathcal{S}(A_0 \otimes A)} \sum_{k=1}^n \|(\text{id}_{A_0} \otimes \Delta_k)(\rho)\|_1.$$

A reference system of input dimension d_A suffices for the diamond norm [6]. The summands are the unconditional, generally subnormalised outcome contributions; no worst conditional-branch normalisation is used.

Remark 2.3 (Operational meaning). *For two instruments $\mathcal{I}, \mathcal{J}$ with equal prior probabilities, the optimal single-use discrimination success probability, with arbitrary reference assistance, is*

$$p_{\text{succ}} = \frac{1}{2} + \frac{1}{4} \|\mathcal{I} - \mathcal{J}\|_{\diamond, \text{inst}},$$

i.e. the advantage over random guessing is $\frac{1}{4} \|\mathcal{I} - \mathcal{J}\|_{\diamond, \text{inst}}$. Thus every radius and width stated in the instrument diamond norm has a direct single-shot discrimination interpretation, up to this conventional factor of $1/4$.

The ambient affine space and its direction space are

$$\begin{aligned} \mathfrak{A}_{A,B}^{(n)} &= \left\{ (\Phi_1, \dots, \Phi_n) : \Phi_k \text{ Hermiticity-preserving, } \sum_{k=1}^n \text{Tr}_B \widehat{J}_{\Phi_k} = \frac{I_{A_0}}{d_A} \right\}, \\ \mathfrak{V}_{A,B}^{(n)} &= \left\{ (\Delta_1, \dots, \Delta_n) : \Delta_k \text{ Hermiticity-preserving, } \sum_{k=1}^n \text{Tr}_B \widehat{J}_{\Delta_k} = 0 \right\}. \end{aligned}$$

Thus only the summed direction is trace-annihilating; individual outcome components need not be. For $\mathcal{I}_0 \in \mathfrak{A}_{A,B}^{(n)}$ define the closed relative instrument-diamond ball

$$\overline{B}_{\diamond, \text{inst}}^{\text{rel}}(\mathcal{I}_0, \rho) = \{ \mathcal{I}_0 + \vec{\Delta} : \vec{\Delta} \in \mathfrak{V}_{A,B}^{(n)}, \|\vec{\Delta}\|_{\diamond, \text{inst}} \leq \rho \},$$

and, for $\mathcal{I}_0 \in \text{Inst}_n(A, B)$, its centred relative in-radius

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_0) = \sup \left\{ \rho \geq 0 : \overline{B}_{\diamond, \text{inst}}^{\text{rel}}(\mathcal{I}_0, \rho) \subseteq \text{Inst}_n(A, B) \right\}.$$

Let $\mathcal{I}_*$ be the uniform depolarising instrument, $(\mathcal{I}_*)_k = \Omega_{A,B}/n$, where $\Omega_{A,B}(X) = \text{Tr}(X)I_B/d_B$, so that $\widehat{\mathcal{J}}_{(\mathcal{I}_*)_k} = I_{A_0 \otimes B}/(nd_A d_B)$.

Remark 2.4 (POVM case $d_B = 1$). When $d_B = 1$, identify $L(B)$ with $\mathbb{C}$. An instrument component is a positive functional $\mathcal{I}_k(X) = \text{Tr}(E_k X)$ for a unique effect $E_k \geq 0$, and the instrument condition is exactly $\sum_{k=1}^n E_k = I_A$. Thus $\text{Inst}_n(A, \mathbb{C})$ is the convex body of n -outcome POVMs on A .

Remark 2.5 (Degenerate singleton case). If $n = d_B = 1$, then $\text{Inst}_1(A, \mathbb{C})$ consists of the unique trace functional $X \mapsto \text{Tr}(X)$: its affine dimension is $D_{A,1}^{(1)} = 0$, its direction space is $\{0\}$, every compression width is 0, and, because every relative ball is the singleton, the centred relative in-radius is $+\infty$ under the supremum convention. Except where explicitly stated, all nontrivial statements in this paper assume

$$nd_B > 1.$$

Remark 2.6 (Exact claims versus certified bounds). The following statements are exact: (i) the affine dimension $D_{A,B}^{(n)}$ (Proposition 3.1); (ii) the centred in-radius at the uniform instrument (Theorem 3.4); (iii) the global covering radius and optimality of $\mathcal{I}_*$ as a centre (Theorem 3.8); (iv) the antipodal profile of the replacement face (Theorem 4.5), which is also the full-body profile when $d_A = 1$; (v) the zero-error Euclidean threshold for continuous encoding with unrestricted decoding (Theorem 5.7); (vi) the diameter (Proposition 3.9); (vii) the constant-code width $\delta_{0, \text{inst}}^\diamond(A, B)$ (Corollary 5.14); (viii) the top-subcritical replacement width (Corollary 5.15); and (ix) the complete width function in the collapse case $nd_B \min\{d_A, d_B\} = 2$ (Theorem 5.16); and (x) the global in-radius over all centres, with $\mathcal{I}_*$ an optimal in-radius centre (Theorem 3.12). The following are certified bounds or open quantities, not exact determinations: (a) the intermediate compression widths $\delta_{r, \text{inst}}^\diamond(A, B)$ for $0 < r < D_{A,B}^{(n)}$ outside the cases (vii)–(ix); (b) the full antipodal profile beyond the certified constructions; (c) optimality of the certified upper envelope $U_r^{(n)}$.

Remark 2.7 (Notation). To avoid collisions, ranks are denoted r_R , the number of input pinching blocks by k_{in} , the number of output pinching blocks by ℓ_{out} , the latent dimension by r (or q for general latent spaces), the covering radius by R , input or support subspaces by W , and the reference system in the covering arguments by E .

3 Exact affine and metric parameters

Proposition 3.1 (Affine dimension of the instrument body). Assume $nd_B > 1$. Then

$$\dim_{\mathbb{R}} \text{aff Inst}_n(A, B) = D_{A,B}^{(n)} := d_A^2(nd_B^2 - 1).$$

Proof. An instrument Choi tuple consists of n Hermitian operators on $A_0 \otimes B$, of total real dimension $nd_A^2 d_B^2$. The normalisation constraint

$$\sum_{k=1}^n \text{Tr}_B \widehat{\mathcal{J}}_{\mathcal{I}_k} = \frac{I_{A_0}}{d_A}$$

imposes $d_A^2 = \dim_{\mathbb{R}} \text{Herm}(A_0)$ real affine equations. The linear map $(X_1, \dots, X_n) \mapsto \sum_k \text{Tr}_B X_k$ from $\bigoplus_k \text{Herm}(A_0 \otimes B)$ to $\text{Herm}(A_0)$ is surjective: for $H \in \text{Herm}(A_0)$ set $X_1 = H \otimes I_B/d_B$ (so $\text{Tr}_B X_1 = H$) and $X_{k \geq 2} = 0$. Hence the constraints are independent. The tuple of $\mathcal{I}_*$ has every

block $\widehat{J}_{(\mathcal{I}_*)_k} = I_{A_0 \otimes B} / (nd_A d_B)$ positive definite; every sufficiently small Hermitian perturbation in $\mathfrak{V}_{A,B}^{(n)}$ therefore remains positive semidefinite. Hence $\mathcal{I}_*$ is a relative interior point, positivity introduces no further affine equalities, and the affine count $nd_A^2 d_B^2 - d_A^2 = d_A^2 (nd_B^2 - 1)$ is attained. $\square$

Lemma 3.2 (Instrument spectral bound). *Let $s = \min\{d_A, d_B\}$ and $\vec{\Delta} \in \mathfrak{V}_{A,B}^{(n)}$. Let*

$$X = \widehat{J}_{\vec{\Delta}} \in \text{Herm}(A_0 \otimes B \otimes C_n), \quad X = \sum_{k=1}^n X_k \otimes |k\rangle\langle k|, \quad X_k = \widehat{J}_{\Delta_k},$$

be the normalised Choi operator of the flagged map $\widehat{\Delta} : A \rightarrow B \otimes C_n$. Then

$$\max\{0, \lambda_{\max}(-X)\} \leq \frac{s}{2d_A} \|\vec{\Delta}\|_{\diamond, \text{inst}}.$$

Proof. If $\lambda_{\max}(-X) \leq 0$ there is nothing to prove. Otherwise, since X is block diagonal, $\lambda_{\max}(-X) = \max_k \lambda_{\max}(-X_k)$. Choose k_0 attaining the maximum and a unit vector $|v\rangle \in A_0 \otimes B$ with $X_{k_0}|v\rangle = -\lambda|v\rangle$, $\lambda = \lambda_{\max}(-X)$. Let $W \subseteq A_0$ be the support of $\text{Tr}_B |v\rangle\langle v|$ and put $b = \dim W$: the rank of $\text{Tr}_B |v\rangle\langle v|$ is the Schmidt rank of $|v\rangle$ across $A_0 \otimes B$, so $b \leq \min\{d_A, d_B\} = s$. If $|e_1\rangle, \dots, |e_b\rangle$ is an orthonormal basis of W and $|\bar{e}_j\rangle \in A$ its coordinatewise conjugate under the fixed Choi bases, define

$$|\omega_W\rangle = \frac{1}{\sqrt{b}} \sum_{j=1}^b |e_j\rangle_{A_0} |\bar{e}_j\rangle_A.$$

With P_W the projection onto W , the normalised-Choi convention gives

$$Y := (\text{id}_{A_0} \otimes \widehat{\Delta})(|\omega_W\rangle\langle\omega_W|) = \frac{d_A}{b} (P_W \otimes I_{B \otimes C_n}) X (P_W \otimes I_{B \otimes C_n}).$$

The direction constraint implies

$$\text{Tr } Y = \frac{d_A}{b} \text{Tr} \left[P_W \sum_{k=1}^n \text{Tr}_B X_k \right] = 0,$$

so Y is Hermitian and traceless. Since $|v\rangle \in W \otimes B$, testing on $|v, k_0\rangle := |v\rangle \otimes |k_0\rangle$ gives

$$\langle v, k_0 | Y | v, k_0 \rangle = -\frac{d_A}{b} \lambda.$$

For a traceless Hermitian operator $Y = Y_+ - Y_-$ one has $\text{Tr } Y_+ = \text{Tr } Y_- = \frac{1}{2} \|Y\|_1$ and $\langle w | Y | w \rangle \geq -\text{Tr } Y_-$ for every unit vector $|w\rangle$, because $Y \geq -Y_-$ and $\langle w | Y_- | w \rangle \leq \text{Tr } Y_-$. Consequently

$$\|Y\|_1 \geq -2\langle v, k_0 | Y | v, k_0 \rangle = \frac{2d_A}{b} \lambda \geq \frac{2d_A}{s} \lambda.$$

The state $|\omega_W\rangle\langle\omega_W|$ is admissible in the instrument diamond-norm optimisation, so $\|\vec{\Delta}\|_{\diamond, \text{inst}} \geq \|Y\|_1$, and rearranging proves the claim. $\square$

Theorem 3.3 (Instrument diamond ball). *Let $s = \min\{d_A, d_B\}$. Then*

$$\overline{B}_{\diamond, \text{inst}}^{\text{rel}} \left(\mathcal{I}_*, \frac{2}{nd_B s} \right) \subseteq \text{Inst}_n(A, B).$$

Proof. Every eigenvalue of the flagged Choi operator of $\mathcal{I}_*$ equals $1/(nd_A d_B)$. If $\vec{\Delta} \in \mathfrak{V}_{A,B}^{(n)}$ with $\|\vec{\Delta}\|_{\diamond, \text{inst}} \leq 2/(nd_B s)$ and $X = \hat{J}_{\vec{\Delta}}$, Lemma 3.2 gives

$$\lambda_{\max}(-X) \leq \frac{s}{2d_A} \cdot \frac{2}{nd_B s} = \frac{1}{nd_A d_B} = \lambda_{\min}(\hat{J}_{\mathcal{I}_*}),$$

so the full flagged Choi operator $\hat{J}_{\mathcal{I}_*} + X$ is positive. Since X is block diagonal this is equivalent to positivity of every outcome block, so each perturbed component is completely positive. The summed marginal is unchanged by the direction constraint, so the sum is trace preserving; and because the blocks are positive with $\sum_k \text{Tr}_B(\hat{J}_{(\mathcal{I}_*)_k} + X_k) = I_{A_0}/d_A$, each partial trace is bounded by I_{A_0}/d_A , so each component is trace-nonincreasing. Hence $\mathcal{I}_* + \vec{\Delta} \in \text{Inst}_n(A, B)$. $\square$

Theorem 3.4 (Exact centred instrument radius). *Assume $nd_B > 1$, and let $s = \min\{d_A, d_B\}$. Then*

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_*) = \frac{2}{nd_B s}.$$

Proof. Theorem 3.3 gives the lower bound. For the reverse inequality it suffices to exhibit a unit-norm direction reaching the boundary at the stated distance. The condition $nd_B > 1$ yields exactly two cases.

Case $d_B > 1$. Choose an s -dimensional input subspace $W_A \subseteq A$ and two isometries $V_0, V_1 : W_A \rightarrow B$ with $\text{Tr}(V_0^\dagger V_1) = 0$ (if $s = 1$, two orthogonal output unit vectors; if $s \geq 2$, fix V_0 , take a trace-zero unitary U on W_A , e.g. diagonal with roots of unity, and set $V_1 = V_0 U$). With P_{W_A} the input projection define the completely positive, trace-nonincreasing maps

$$\mathcal{E}_j(Z) = V_j P_{W_A} Z P_{W_A} V_j^\dagger, \quad j = 0, 1.$$

Choose an orthonormal basis $|e_1\rangle, \dots, |e_s\rangle$ of W_A and set

$$|\omega_{W_A}\rangle = \frac{1}{\sqrt{s}} \sum_{a=1}^s |\bar{e}_a\rangle_{A_0} |e_a\rangle_A, \quad P_j = (\text{id} \otimes \mathcal{E}_j)(|\omega_{W_A}\rangle \langle \omega_{W_A}|).$$

The P_j are rank-one projectors with

$$\text{Tr}_B P_0 = \text{Tr}_B P_1 = \frac{P_{\bar{W}_A}}{s}, \quad \text{Tr}(P_0 P_1) = \left| \frac{1}{s} \text{Tr}(V_0^\dagger V_1) \right|^2 = 0,$$

so the maximally entangled input produces orthogonal pure outputs and $\|\mathcal{E}_1 - \mathcal{E}_0\|_\diamond = 2$. Define the instrument direction

$$\Delta_1 = \frac{1}{2}(\mathcal{E}_1 - \mathcal{E}_0), \quad \Delta_k = 0 \ (k \geq 2).$$

Since $\text{Tr} \mathcal{E}_1(Z) = \text{Tr}(P_{W_A} Z) = \text{Tr} \mathcal{E}_0(Z)$, the summed direction is trace-annihilating, so $\vec{\Delta} \in \mathfrak{V}_{A,B}^{(n)}$; its flagged diamond norm is 1 and its first normalised Choi block is $\hat{J}_{\Delta_1} = \frac{s}{2d_A}(P_1 - P_0)$. At $t = 2/(nd_B s)$ the first block of $\mathcal{I}_* + t\vec{\Delta}$ equals

$$\frac{1}{nd_A d_B} (I_{A_0 \otimes B} + P_1 - P_0),$$

which is positive, with the unit vector supporting P_0 a zero eigenvector because $P_0 P_1 = 0$. The remaining blocks are unchanged and the marginal constraint is preserved because $\text{Tr}_B P_0 = \text{Tr}_B P_1$. Hence the ray reaches the relative boundary at distance t . For $t' > t$ the eigenvalue on the support of P_0 becomes $1/(nd_A d_B) - t's/(2d_A) < 0$, so $\mathcal{I}_* + t'\vec{\Delta}$ is not an instrument. No closed relative ball of radius larger than t is contained in $\text{Inst}_n(A, B)$.

Case $d_B = 1$, $n \geq 2$ (POVMs). Choose a rank-one projection P on A and define the positive functional $\mathcal{F} : L(A) \rightarrow \mathbb{C}$, $\mathcal{F}(Z) = \text{Tr}(PZ)$. Define instrument directions as scalar-output maps

$$\Delta_1 = -\frac{1}{2}\mathcal{F}, \quad \Delta_2 = +\frac{1}{2}\mathcal{F}, \quad \Delta_k = 0 \ (k \geq 3),$$

so that $\Delta_1 + \Delta_2 = 0$ as linear functionals and the summed direction vanishes. For every reference-assisted input state ξ , set

$$\zeta_E = \text{Tr}_A[(I_E \otimes P)\xi(I_E \otimes P)] = \text{Tr}_A[(I_E \otimes P)\xi] \geq 0.$$

Then

$$(\text{id}_E \otimes \widehat{\vec{\Delta}})(\xi) = \frac{1}{2} \zeta_E \otimes (|2\rangle\langle 2| - |1\rangle\langle 1|),$$

whose trace norm is $\text{Tr} \zeta_E \leq 1$, with equality attained by an input supported on P ; hence $\|\vec{\Delta}\|_{\diamond, \text{inst}} = 1$. In the fixed Choi basis, $\widehat{\vec{\mathcal{F}}} = P^\top/d_A$. At $t = 2/n$ the first two Choi blocks of $\mathcal{I}_* + t\vec{\Delta}$ are

$$\frac{I_{A_0} - P^\top}{nd_A}, \quad \frac{I_{A_0} + P^\top}{nd_A},$$

all other blocks remaining $I_{A_0}/(nd_A)$. These blocks are positive, the first is singular, and their sum is I_{A_0}/d_A ; the ray reaches the POVM boundary at distance $2/n$, which is the stated formula for $d_B = s = 1$. For $t' > 2/n$ the first block has the negative eigenvalue $1/(nd_A) - t'/(2d_A)$ on the range of $P^\top$, proving maximality. $\square$

Remark 3.5 (Physical origin of the radius). *Each Choi block of the uniform instrument has positivity margin $1/(nd_A d_B)$, while a unit instrument-diamond perturbation can create a negative Choi eigenvalue of size at most $s/(2d_A)$; their ratio is the exact radius $2/(nd_B s)$. The factor $1/n$ reflects the distribution of the uniform trace budget among n outcomes at the uniform centre and is not a universal $1/n$ law for arbitrary centres or arbitrary local robustness notions. The same numerical positivity-budget calculation is realised by two distinct witnesses living in different faces of the positive cone: an orthogonal-output isometry witness when $d_B > 1$, and an effect-transfer witness (a rank-one measurement effect moved between two classical outcomes) when $d_B = 1$, $n \geq 2$.*

Remark 3.6 (Global in-radius). *Theorem 3.4 determines the largest ball centred at $\mathcal{I}_*$; whether some other centre of $\text{Inst}_n(A, B)$ could carry a larger relative ball — which would improve the intermediate certified bounds below — is settled in Theorem 3.12 below: no centre does, and the global in-radius*

$$\sup_{\mathcal{I}_0 \in \text{Inst}_n(A, B)} \rho_{\diamond, \text{inst}}(\mathcal{I}_0)$$

equals the centred value $2/(nd_B \min\{d_A, d_B\})$.

Lemma 3.7 (Bipartite domination for instrument blocks). *Let $\sigma_{EB} \geq 0$ be subnormalised, with $\rho_E = \text{Tr}_B \sigma_{EB}$ and $r_R = \text{rank } \rho_E$. Then*

$$\sigma_{EB} \leq \min\{r_R, d_B\} \rho_E \otimes I_B.$$

Proof. If $\rho_E = 0$ there is nothing to prove. Otherwise let $E_0 = \text{supp } \rho_E$, so that $\rho_E^{-1/2}$ is defined on E_0 (extended by zero on the orthocomplement), and set

$$Z = (\rho_E^{-1/2} \otimes I_B) \sigma_{EB} (\rho_E^{-1/2} \otimes I_B).$$

Then $Z \geq 0$ and $\text{Tr}_B Z = I_{E_0}$. Write $Z = \sum_\alpha |K_\alpha\rangle\rangle\langle\langle K_\alpha|$ with $\sum_\alpha K_\alpha^\dagger K_\alpha = I_{E_0}$ in the vectorisation convention $|K\rangle\rangle = \sum_i |i\rangle \otimes K|i\rangle$. For a unit vector $|v\rangle \in E_0 \otimes B$ let $V : E_0 \rightarrow B$

be the corresponding matrix of rank $b \leq \min\{r_R, d_B\}$. Using $|\text{Tr } M|^2 \leq \text{rank}(M) \|M\|_2^2$ with $M = K_\alpha V^\dagger$ of rank at most b ,

$$\langle v | Z | v \rangle = \sum_{\alpha} |\text{Tr}(V^\dagger K_\alpha)|^2 \leq b \sum_{\alpha} \text{Tr}(V K_\alpha^\dagger K_\alpha V^\dagger) = b.$$

Thus $Z \leq \min\{r_R, d_B\} I_{E_0 \otimes B}$, and conjugation by $\rho_E^{1/2} \otimes I_B$ proves the claim. $\square$

Theorem 3.8 (Exact instrument covering radius). *Let $s = \min\{d_A, d_B\}$. For $\Psi \in \mathfrak{A}_{A,B}^{(n)}$ define*

$$R_{\text{inst}}(\Psi) = \sup_{\mathcal{I} \in \text{Inst}_n(A,B)} \|\mathcal{I} - \Psi\|_{\diamond, \text{inst}}.$$

Then the uniform depolarising instrument is an optimal centre over all of $\mathfrak{A}_{A,B}^{(n)}$, and

$$R_{\text{inst}}(\mathcal{I}_*) = 2 \left(1 - \frac{1}{nd_B s} \right) = 2 - \rho_{\diamond, \text{inst}}(\mathcal{I}_*).$$

Consequently the exact constant-code instrument width equals this value.

Proof. Fix a reference-assisted input state and write the flagged output of an instrument as

$$\sigma_{EBC_n} = \sum_{k=1}^n \sigma_{k,EB} \otimes |k\rangle\langle k|, \quad \sum_k \text{Tr}_B \sigma_{k,EB} = \rho_E.$$

Put $\rho_{k,E} = \text{Tr}_B \sigma_{k,EB}$. Since $\text{rank } \rho_{k,E} \leq d_A$, Lemma 3.7 gives

$$\sigma_{k,EB} \leq s \rho_{k,E} \otimes I_B \leq s \rho_E \otimes I_B.$$

Therefore

$$\sigma_{EBC_n} \leq nd_B s \left(\rho_E \otimes \frac{I_B}{d_B} \otimes \frac{I_{C_n}}{n} \right),$$

and the state in parentheses is the output of $\mathcal{I}_*$. If states satisfy $\sigma \leq K\tau$, let P be the positive spectral projection of $\sigma - \tau$ and put $a = \text{Tr}(P\sigma)$, $b = \text{Tr}(P\tau)$. Then $a \leq Kb$ and $a \leq 1$, so

$$\frac{1}{2} \|\sigma - \tau\|_1 = a - b \leq a \left(1 - \frac{1}{K} \right) \leq 1 - \frac{1}{K}.$$

Taking the supremum over inputs yields $R_{\text{inst}}(\mathcal{I}_*) \leq 2(1 - 1/(nd_B s))$.

For equality, choose an s -dimensional input subspace $W \subseteq A$, an isometry $V : W \rightarrow B$, and a state ω on B . With P_W the subspace projection define the channel

$$\Phi(X) = V P_W X P_W V^\dagger + \text{Tr}[(I_A - P_W)X] \omega,$$

whose action on the selected Schmidt-rank- s subspace is a pure isometry while its action outside W is a fixed replacement (in particular, no claim of extremality is made or needed). Set $\mathcal{I}_1 = \Phi$ and $\mathcal{I}_k = 0$ for $k \geq 2$. A maximally entangled input of Schmidt rank s on W gives a pure output in the first flag, whereas $\mathcal{I}_*$ gives the maximally mixed state on the corresponding $nd_B s$ -dimensional support; the trace norm of their difference is $2(1 - 1/(nd_B s))$ (equivalently, the trace distance is $1 - 1/(nd_B s)$). Hence $R_{\text{inst}}(\mathcal{I}_*) \geq 2(1 - 1/(nd_B s))$, and equality holds.

Optimality of the centre. Let $G = \text{U}(B) \times S_n$ act on $\mathfrak{A}_{A,B}^{(n)}$ by

$$[(U, \pi) \cdot \Psi]_k = \mathcal{U}_U \circ \Psi_{\pi^{-1}(k)}, \quad \mathcal{U}_U(X) = UXU^\dagger.$$

This action preserves $\text{Inst}_n(A, B)$ and the instrument diamond metric, hence $R_{\text{inst}}(g \cdot \Psi) = R_{\text{inst}}(\Psi)$. The functional R_{inst} is convex, being a supremum of norms. Let $\bar{\Psi}$ be the average

of Ψ over G with respect to Haar measure on $U(B)$ and the uniform measure on S_n . Jensen's inequality gives $R_{\text{inst}}(\bar{\Psi}) \leq R_{\text{inst}}(\Psi)$. Averaging first over $U(B)$, Schur's lemma [6] gives for each component

$$\int_{U(B)} U \Psi_k(X) U^\dagger dU = \frac{\text{Tr}(\Psi_k(X))}{d_B} I_B;$$

averaging then over outcome permutations, the k -th component becomes

$$\frac{1}{n} \sum_{j=1}^n \frac{\text{Tr}(\Psi_j(X))}{d_B} I_B = \frac{\text{Tr}(X)}{nd_B} I_B,$$

because the summed map is trace preserving. Thus $\bar{\Psi} = \mathcal{I}_*$, and $R_{\text{inst}}(\mathcal{I}_*) \leq R_{\text{inst}}(\Psi)$ for every $\Psi \in \mathfrak{A}_{A,B}^{(n)}$. Since $\mathcal{I}_* \in \text{Inst}_n(A, B)$, the constant-code width equals the covering radius about $\mathcal{I}_*$. $\square$

Proposition 3.9 (Exact instrument diameter). *Assume $nd_B > 1$. Then*

$$\text{diam}_{\diamond, \text{inst}}(\text{Inst}_n(A, B)) = 2.$$

Proof. For any two instruments $\mathcal{I}, \mathcal{J}$, the flagged maps $\hat{\mathcal{I}}, \hat{\mathcal{J}}$ are channels, so

$$\|\mathcal{I} - \mathcal{J}\|_{\diamond, \text{inst}} = \|\hat{\mathcal{I}} - \hat{\mathcal{J}}\|_{\diamond} \leq \|\hat{\mathcal{I}}\|_{\diamond} + \|\hat{\mathcal{J}}\|_{\diamond} = 2.$$

Equality is achieved by replacement instruments with orthogonal flagged states, which exist whenever $nd_B > 1$: if $d_B \geq 2$, take two orthogonal output states in the same flag; if $d_B = 1$, then $n \geq 2$, and take two orthogonal flags. By the replacement isometry (Lemma 4.4) the corresponding instruments are at distance 2. $\square$

Corollary 3.10 (Metric complementarity of the uniform instrument). *Assume $nd_B > 1$. Then*

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_*) + R_{\text{inst}}(\mathcal{I}_*) = 2 = \text{diam}_{\diamond, \text{inst}}(\text{Inst}_n(A, B)).$$

This is a numerical complementarity identity for the centred in-radius and the covering radius about $\mathcal{I}_$. It does not assert that the in-radius witness and the covering-radius witness lie on a common extremal chord through $\mathcal{I}_*$; the two witnesses are constructed independently and their collinearity is not claimed.*

Proof. By Theorems 3.4 and 3.8,

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_*) + R_{\text{inst}}(\mathcal{I}_*) = \frac{2}{nd_B s} + 2 \left(1 - \frac{1}{nd_B s}\right) = 2,$$

and $\text{diam}_{\diamond, \text{inst}}(\text{Inst}_n(A, B)) = 2$ by Proposition 3.9. The identity is purely algebraic: the two summands are witnessed by different, independently constructed points of the body. $\square$

Lemma 3.11 (Concavity and Lipschitz continuity of the in-radius). *Write*

$$B_{\mathfrak{B}}(\rho) = \{\vec{\Delta} \in \mathfrak{B}_{A,B}^{(n)} : \|\vec{\Delta}\|_{\diamond, \text{inst}} \leq \rho\}$$

for the closed direction ball. For centres $\mathcal{I}_0, \mathcal{I}_1 \in \text{Inst}_n(A, B)$ and $\lambda \in [0, 1]$,

$$\rho_{\diamond, \text{inst}}(\lambda \mathcal{I}_0 + (1 - \lambda) \mathcal{I}_1) \geq \lambda \rho_{\diamond, \text{inst}}(\mathcal{I}_0) + (1 - \lambda) \rho_{\diamond, \text{inst}}(\mathcal{I}_1),$$

and

$$|\rho_{\diamond, \text{inst}}(\mathcal{I}_0) - \rho_{\diamond, \text{inst}}(\mathcal{I}_1)| \leq \|\mathcal{I}_0 - \mathcal{I}_1\|_{\diamond, \text{inst}}.$$

In particular, the in-radius is a continuous concave function of the centre on the body.

Proof. For centrally symmetric convex balls one has the Minkowski identity

$$\lambda B_{\mathfrak{B}}(\rho_0) + (1 - \lambda) B_{\mathfrak{B}}(\rho_1) = B_{\mathfrak{B}}(\lambda \rho_0 + (1 - \lambda) \rho_1) :$$

the inclusion “ $\subseteq$ ” is the triangle inequality, and “ $\supseteq$ ” follows from the decomposition $\rho u = \lambda \rho_0 u + (1 - \lambda) \rho_1 u$ with $\rho = \lambda \rho_0 + (1 - \lambda) \rho_1$ and $\|u\|_{\diamond, \text{inst}} \leq 1$. Assume $\mathcal{I}_i + B_{\mathfrak{B}}(\rho_i) \subseteq \text{Inst}_n(A, B)$; every point of $\lambda \mathcal{I}_0 + (1 - \lambda) \mathcal{I}_1 + \lambda B_{\mathfrak{B}}(\rho_0) + (1 - \lambda) B_{\mathfrak{B}}(\rho_1)$ is a convex combination of two instruments, hence an instrument. Optimising ρ_0 and ρ_1 independently gives the concavity.

For the Lipschitz bound, let $\delta = \|\mathcal{I}_0 - \mathcal{I}_1\|_{\diamond, \text{inst}}$, so $\mathcal{I}_1 - \mathcal{I}_0 \in B_{\mathfrak{B}}(\delta)$. If $\rho_{\diamond, \text{inst}}(\mathcal{I}_0) \leq \delta$ there is nothing to prove. Otherwise, with $\rho = \rho_{\diamond, \text{inst}}(\mathcal{I}_0)$,

$$\mathcal{I}_1 + B_{\mathfrak{B}}(\rho - \delta) = \mathcal{I}_0 + (\mathcal{I}_1 - \mathcal{I}_0) + B_{\mathfrak{B}}(\rho - \delta) \subseteq \mathcal{I}_0 + B_{\mathfrak{B}}(\delta) + B_{\mathfrak{B}}(\rho - \delta) = \mathcal{I}_0 + B_{\mathfrak{B}}(\rho) \subseteq \text{Inst}_n(A, B),$$

so $\rho_{\diamond, \text{inst}}(\mathcal{I}_1) \geq \rho_{\diamond, \text{inst}}(\mathcal{I}_0) - \delta$; exchanging the two centres gives the absolute value. $\square$

Theorem 3.12 (Global in-radius). *Assume $nd_B > 1$. Then the uniform depolarising instrument is an optimal in-radius centre, and*

$$\sup_{\mathcal{I}_0 \in \text{Inst}_n(A, B)} \rho_{\diamond, \text{inst}}(\mathcal{I}_0) = \rho_{\diamond, \text{inst}}(\mathcal{I}_*) = \frac{2}{nd_B \min\{d_A, d_B\}}.$$

Proof. Let $G = \text{U}(B) \times S_n$ act on $\mathfrak{A}_{A, B}^{(n)}$ as in the proof of Theorem 3.8, and let μ be the normalised Haar measure. The action restricts to the direction space: for $\vec{\Delta} \in \mathfrak{B}_{A, B}^{(n)}$, the components of $g \cdot \vec{\Delta}$ are $\mathcal{U}_U \circ \Delta_{\pi^{-1}(k)}$, and the A_0 -marginal of a Choi operator is invariant under unitary conjugation of B ,

$$\text{Tr}_B \hat{J}_{\mathcal{U}_U \circ \Delta} = \text{Tr}_B [(I_{A_0} \otimes U) \hat{J}_{\Delta} (I_{A_0} \otimes U^\dagger)] = \text{Tr}_B \hat{J}_{\Delta},$$

so the summed marginal of $g \cdot \vec{\Delta}$ still vanishes. With W_π the flag-permutation unitary on C_n ($W_\pi|j\rangle = |\pi(j)\rangle$), the flagged operator of $g \cdot \vec{\Delta}$ satisfies

$$\widehat{g \cdot \vec{\Delta}}(X) = (U \otimes W_\pi) \widehat{\vec{\Delta}}(X) (U^\dagger \otimes W_\pi^\dagger) \quad (X \in \text{L}(A)),$$

that is, $\widehat{g \cdot \vec{\Delta}} = \mathcal{U}_{U \otimes W_\pi} \circ \widehat{\vec{\Delta}}$ is the post-composition of $\widehat{\vec{\Delta}}$ by the unitary channel of $U \otimes W_\pi$ on $B \otimes C_n$; post-composition by unitary channels preserves the diamond norm. Hence g maps $B_{\mathfrak{B}}(\rho)$ onto itself and $\text{Inst}_n(A, B)$ onto itself, and consequently

$$\rho_{\diamond, \text{inst}}(g \cdot \mathcal{I}_0) = \rho_{\diamond, \text{inst}}(\mathcal{I}_0) \quad (g \in G, \mathcal{I}_0 \in \text{Inst}_n(A, B)).$$

By the Schur-lemma and permutation averages computed in the proof of Theorem 3.8 — valid verbatim for every point of $\mathfrak{A}_{A, B}^{(n)}$ — the barycentre of every orbit is $\mathcal{I}_*$:

$$\int_G g \cdot \mathcal{I}_0 d\mu(g) = \mathcal{I}_*.$$

The orbit map $g \mapsto g \cdot \mathcal{I}_0$ is continuous, and the in-radius is concave and continuous on the body (Lemma 3.11); Jensen’s inequality (the integral being a limit of convex combinations of orbit points, to which concavity extends by continuity) gives

$$\rho_{\diamond, \text{inst}}(\mathcal{I}_*) = \rho_{\diamond, \text{inst}}\left(\int_G g \cdot \mathcal{I}_0 d\mu(g)\right) \geq \int_G \rho_{\diamond, \text{inst}}(g \cdot \mathcal{I}_0) d\mu(g) = \rho_{\diamond, \text{inst}}(\mathcal{I}_0).$$

Together with the exact centred value of Theorem 3.4, this proves the claim. $\square$

Remark 3.13 (Concave–convex duality of the two radii). *Theorem 3.12 is the mirror image of the covering-radius symmetrisation in Theorem 3.8. The covering radius is a convex function of the centre, and Jensen’s inequality at the Haar barycentre $\mathcal{I}_*$ shows that no centre has a smaller covering radius; the in-radius is a concave function of the centre (Lemma 3.11), and the same Jensen inequality shows that no centre has a larger in-radius. The uniform depolarising instrument is thus an optimal centre for both radii simultaneously, and the complementarity identity of Corollary 3.10 holds at this common optimal centre. Two by-products are worth recording. First, the ball term $\rho_n = 2/(nd_{Bs})$ of the certified intermediate lower bound $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$ is optimal among all centred-ball certificates: re-centring the radial construction cannot certify more than ρ_n at any centre (compare Remark 5.20(iii)). Second, the Lipschitz bound of Lemma 3.11 is a robustness statement: every centre within distance ε of $\mathcal{I}_*$ still carries a relative ball of radius $\rho_n - \varepsilon$. Whether $\mathcal{I}_*$ is the unique in-radius maximiser is not addressed here.*

4 Antipodal certificates and the lower envelope

Definition 4.1 (Antipodal profile). *For any nonempty metric space (K, d) define*

$$\alpha_r(K) = \frac{1}{2} \sup_{h \in C(S^r, K)} \min_{u \in S^r} d(h(u), h(-u)),$$

with the instrument diamond metric when $K \subseteq \text{Inst}_n(A, B)$.

Remark 4.2 (Terminology for the four radius notions). *Four distinct radius-like quantities are used throughout this paper, always written with their qualifiers: (i) the centred relative in-radius $\rho_{\diamond, \text{inst}}(\mathcal{I}_0)$, the largest relative ball about a fixed centre that is contained in the body; (ii) the worst-case covering radius $R_{\text{inst}}(\Psi)$, the maximal distance from a centre to the body; (iii) the antipodal profile α_r , a width-like sequence indexed by the sphere dimension r , which is not the radius of any ball; and (iv) the global in-radius, the supremum of the centred in-radius over all centres of the body (Theorem 3.12). The word “radius” without a qualifier is avoided.*

Proposition 4.3 (Monotonicity of profiles, widths, and envelopes). *For every nonempty metric space K ,*

$$\alpha_{r+1}(K) \leq \alpha_r(K) \quad (r \geq 0),$$

and if K is compact then $0 \leq \alpha_r(K) \leq \frac{1}{2} \text{diam}(K)$. Consequently, if $\alpha_D(K) = 0$ then $\alpha_r(K) = 0$ for all $r \geq D$. The Euclidean widths satisfy

$$\delta_{r+1, \text{inst}}^\diamond(A, B) \leq \delta_{r, \text{inst}}^\diamond(A, B)$$

by padding the encoder with a zero coordinate. The certified lower envelope $\beta_r^{\text{best}, (n)}$ is non-increasing by construction, and the certified upper envelope $U_r^{(n)}$ is non-increasing because increasing r only enlarges the set of admissible upper bounds.

Proof. Let $h : S^{r+1} \rightarrow K$ be continuous and restrict it to an equatorial copy $S^r \simeq \{x \in S^{r+1} : x_{r+1} = 0\}$. The antipodal map on the equator is the ordinary antipodal map on S^r , hence

$$\min_{u \in S^r} d(h(u), h(-u)) \geq \min_{u \in S^{r+1}} d(h(u), h(-u)).$$

Taking one half and the supremum over all continuous $h : S^{r+1} \rightarrow K$ gives $\alpha_r(K) \geq \alpha_{r+1}(K)$. The diameter bound is immediate, and the remaining claims are explained in the statement. $\square$

Let

$$\mathcal{S}_{\text{flag}}(B \otimes C_n) = \left\{ \tau = \sum_{k=1}^n \tau_k \otimes |k\rangle\langle k| : \tau_k \geq 0, \sum_{k=1}^n \text{Tr} \tau_k = 1 \right\}.$$

For $\tau \in \mathcal{S}_{\text{flag}}(B \otimes C_n)$ define the replacement instrument $\vec{\mathcal{R}}_\tau$ by $(\vec{\mathcal{R}}_\tau)_k(X) = \text{Tr}(X)\tau_k$; each component is completely positive and trace-nonincreasing and the sum is a channel. Write

$$\text{RepInst}_n(A, B) = \{ \vec{\mathcal{R}}_\tau : \tau \in \mathcal{S}_{\text{flag}}(B \otimes C_n) \}.$$

The set $\text{RepInst}_n(A, B)$ is a face of $\text{Inst}_n(A, B)$: if $\lambda \vec{\mathcal{R}}_\tau + (1 - \lambda)\mathcal{J} \in \text{RepInst}_n(A, B)$ with $\mathcal{J} \in \text{Inst}_n(A, B)$ and $0 < \lambda < 1$, write $\lambda \vec{\mathcal{R}}_\tau + (1 - \lambda)\mathcal{J} = \vec{\mathcal{R}}_\sigma$; then each component of $\mathcal{J}$ satisfies $\mathcal{J}_k(X) = \text{Tr}(X)\tilde{\tau}_k$ with $\tilde{\tau}_k = (\sigma_k - \lambda\tau_k)/(1 - \lambda)$, complete positivity of $\mathcal{J}$ forces $\tilde{\tau}_k \geq 0$, and the summed trace condition gives $\sum_k \text{Tr} \tilde{\tau}_k = 1$, so $\mathcal{J} = \vec{\mathcal{R}}_{\tilde{\tau}} \in \text{RepInst}_n(A, B)$. This justifies the name *replacement face* used below.

Lemma 4.4 (Replacement-instrument isometry). *For all flagged states τ, σ ,*

$$\|\vec{\mathcal{R}}_\tau - \vec{\mathcal{R}}_\sigma\|_{\diamond, \text{inst}} = \|\tau - \sigma\|_1 = \sum_{k=1}^n \|\tau_k - \sigma_k\|_1.$$

Proof. The flagged channel of $\vec{\mathcal{R}}_\tau$ is the replacement channel $X \mapsto \text{Tr}(X)\tau$. For every reference-assisted input state ξ ,

$$(\text{id} \otimes (\widehat{\vec{\mathcal{R}}_\tau} - \widehat{\vec{\mathcal{R}}_\sigma}))(\xi) = \text{Tr}_A(\xi) \otimes (\tau - \sigma),$$

and $\text{Tr}_A(\xi)$ is a state, so the trace norm equals $\|\tau - \sigma\|_1$ by multiplicativity; the supremum over inputs is unnecessary. The second equality follows from block diagonality in the flag. $\square$

Theorem 4.5 (Perfectly separated replacement instruments). *Assume $nd_B > 1$. Flagged replacement instruments contain a continuous image of $S^{nd_B^2 - 2}$ whose antipodes have instrument-diamond distance two, and consequently*

$$\alpha_r(\text{Inst}_n(A, B)) = 1 \quad (0 \leq r \leq nd_B^2 - 2).$$

For the replacement family itself the profile is exact:

$$\alpha_r(\text{RepInst}_n(A, B)) = \begin{cases} 1, & 0 \leq r \leq nd_B^2 - 2, \\ 0, & r \geq nd_B^2 - 1. \end{cases}$$

Proof. The real vector space of flag-block-diagonal Hermitian operators on $B \otimes C_n$ has dimension nd_B^2 ; its trace-zero subspace

$$\mathfrak{H}_{\text{flag}, 0} = \left\{ X = \sum_{k=1}^n X_k \otimes |k\rangle\langle k| : X_k \in \text{Herm}(B), \sum_k \text{Tr} X_k = 0 \right\}$$

has dimension $m := nd_B^2 - 1$. Choose any Euclidean norm and let $S(\mathfrak{H}_{\text{flag}, 0}) \cong S^{m-1} = S^{nd_B^2 - 2}$ be its unit sphere. For X on the sphere write the Jordan decomposition $X = X_+ - X_-$; functional calculus preserves block diagonality and makes $X \mapsto X_+$ continuous. Since X is nonzero and traceless, $t_X := \text{Tr} X_+ = \text{Tr} X_- > 0$. The map $h(X) = X_+/t_X$ is a continuous map into $\mathcal{S}_{\text{flag}}(B \otimes C_n)$ — a continuous image of the sphere, *not claimed to be an embedding*, which suffices for antipodal-width arguments — and $h(-X) = X_-/t_X$ has orthogonal support, so $\|h(X) - h(-X)\|_1 = 2$. Composing with the replacement embedding of Lemma 4.4 gives the claimed sphere of instruments; equatorial restriction supplies every smaller index, and the reverse inequality follows from the diameter bound (Proposition 3.9).

For the replacement family above this range, compose any map $S^r \rightarrow \text{RepInst}_n(A, B)$ with the inverse isometry and affine coordinates in $\mathbb{R}^m$. For $r = m$ Borsuk–Ulam [9] applied to $S^m \rightarrow \text{RepInst}_n(A, B) \rightarrow \mathbb{R}^m$ produces an antipodal collision; for $r > m$, restrict the map to an equatorial $S^m \subset S^r$ first. Every such map therefore has minimum antipodal distance zero, proving the second part of the formula. $\square$

Remark 4.6 (In-radius versus antipodal profile). $\rho_n = 2/(nd_B s)$ certifies a centrally symmetric norm ball in the full direction space $\mathfrak{V}_{A,B}^{(n)}$, while the sphere $S^{nd_B^2-2}$ via $X \mapsto X_+/\text{Tr } X_+$ certifies a non-linear, antipodally 2-separated sphere in the flagged replacement face. There is no contradiction between $\rho_n < 1$ and $\alpha_r = 1$ for $r \leq nd_B^2 - 2$: these are different geometric mechanisms (a ball radius versus an antipodal profile) acting on different subsets of the body. The ball bound governs the intermediate regime, while the replacement sphere governs the small-index regime.

Theorem 4.7 (Full-direction antipodal sphere). Assume $nd_B > 1$, and let $D = D_{A,B}^{(n)}$. For every $0 \leq r \leq D - 1$ there exists a continuous map $h : S^r \rightarrow \text{Inst}_n(A, B)$ such that

$$\|h(u) - h(-u)\|_{\diamond, \text{inst}} \geq \frac{2}{d_A} \quad \forall u \in S^r.$$

Consequently $\alpha_r(\text{Inst}_n(A, B)) \geq 1/d_A$ for all $0 \leq r < D$. When $d_A = 1$ the separation is 2, hence $\alpha_r = 1$ for all $r < D_{1,B}^{(n)}$, consistent with the exact replacement profile.

Proof. Identify instruments with their normalised flagged Choi operators

$$J \in \text{Herm}(A_0 \otimes B \otimes C_n), \quad J \geq 0, \quad \text{Tr}_{B \otimes C_n} J = \frac{I_{A_0}}{d_A},$$

flag-block diagonal in C_n . The direction space

$$\mathcal{V} = \{Z \in \text{Herm}(A_0 \otimes B \otimes C_n)_{\text{flag}} : \text{Tr}_{B \otimes C_n} Z = 0\}$$

has real dimension D . Equip $\mathcal{V}$ with a Euclidean norm and let $S(\mathcal{V}) \cong S^{D-1}$ be its unit sphere; it suffices to construct the map on S^{D-1} , since equatorial restriction supplies every smaller r . Fix the uniform flagged state

$$\tau_0 = \frac{I_B}{d_B} \otimes \frac{I_{C_n}}{n}.$$

For $Z \in S(\mathcal{V})$ write the Jordan decomposition $Z = Z_+ - Z_-$ with $Z_+, Z_- \geq 0$, $Z_+ Z_- = 0$; functional calculus preserves flag-block diagonality. Since $\text{Tr}_{B \otimes C_n} Z = 0$,

$$\text{Tr}_{B \otimes C_n} Z_+ = \text{Tr}_{B \otimes C_n} Z_-.$$

Let $t = \text{Tr } Z_+ = \text{Tr } Z_-$. We claim $t > 0$: if $t = 0$ then $Z_+ = 0$, so $Z = -Z_- \leq 0$; but then $\text{Tr } Z = \text{Tr}_{A_0} \text{Tr}_{B \otimes C_n} Z = 0$ with $Z \leq 0$ forces $Z = 0$, contradicting $Z \in S(\mathcal{V})$. Define the normalised states

$$P = \frac{Z_+}{t}, \quad Q = \frac{Z_-}{t},$$

which have orthogonal supports and share the same A_0 -marginal

$$\mu := \text{Tr}_{B \otimes C_n} P = \text{Tr}_{B \otimes C_n} Q.$$

If $d_A = 1$ the marginal condition is automatic (A_0 is one-dimensional and $\mu = 1$), and we set $J^+(Z) = P$, $J^-(Z) = Q$. Otherwise put $\lambda = 1/d_A$. Since μ is a state on A_0 , $\mu \leq I_{A_0}$, hence $\lambda\mu \leq I_{A_0}/d_A$. Define the correction state

$$\nu = \frac{\frac{I_{A_0}}{d_A} - \lambda\mu}{1 - \lambda} = \frac{I_{A_0} - \mu}{d_A - 1} \geq 0, \quad \text{Tr } \nu = 1,$$

and set

$$J^+(Z) = \lambda P + (1 - \lambda)\nu \otimes \tau_0, \quad J^-(Z) = \lambda Q + (1 - \lambda)\nu \otimes \tau_0.$$

Both $J^\pm(Z)$ are positive and flag-block diagonal with A_0 -marginal

$$\lambda\mu + (1 - \lambda)\nu = \frac{I_{A_0}}{d_A},$$

so each is the normalised flagged Choi operator of an instrument; let $h(Z)$ be the instrument with Choi operator $J^+(Z)$. For $-Z$ the positive and negative parts are interchanged while t and μ are unchanged, so ν is unchanged and $h(-Z)$ is the instrument with Choi operator $J^-(Z)$. The Choi difference is

$$J^+(Z) - J^-(Z) = \lambda(P - Q)$$

(with $\lambda = 1$ when $d_A = 1$), and since P, Q are states with orthogonal supports,

$$\|J^+(Z) - J^-(Z)\|_1 = 2\lambda = \frac{2}{d_A}.$$

For any Hermiticity-preserving tuple $\vec{\Delta}$ the instrument diamond norm dominates the trace norm of its normalised flagged Choi operator, because testing on the maximally entangled input state gives the Choi operator itself. Therefore

$$\|h(Z) - h(-Z)\|_{\diamond, \text{inst}} \geq \frac{2}{d_A}.$$

Finally, $Z \mapsto Z_+$ is continuous on Hermitian operators and $t = \text{Tr } Z_+$ is continuous and strictly positive on the compact unit sphere $S(\mathcal{V})$, hence bounded away from zero there; so $Z \mapsto J^+(Z)$ is continuous. Equatorial restriction proves the theorem for all $0 \leq r \leq D - 1$. $\square$

Remark 4.8 (Tunable mixing parameter). *In the proof of Theorem 4.7, the mixing parameter may be chosen as a function of Z : since $\mu = \mu(Z)$ is a state on A_0 we have $\|\mu\|_\infty \in [1/d_A, 1]$, and $I_{A_0}/d_A - \lambda\mu \geq 0$ holds if and only if $\lambda \leq 1/(d_A\|\mu\|_\infty)$. Taking $\lambda(Z) = 1/(d_A\|\mu(Z)\|_\infty) \in [1/d_A, 1]$ therefore increases the pointwise separation to*

$$\|J^+(Z) - J^-(Z)\|_1 = \frac{2}{d_A\|\mu(Z)\|_\infty} \geq \frac{2}{d_A},$$

with $\lambda = 1$ (no filler needed at all) exactly when $\mu(Z) = I_{A_0}/d_A$. The uniform certificate $\alpha_r \geq 1/d_A$ is unchanged; we fixed $\lambda = 1/d_A$ in the proof to keep continuity of the construction transparent, since near points with $\mu(Z) = I_{A_0}/d_A$ the variable quotient $\nu(Z)$ would require a separate gluing discussion.

Corollary 4.9 (Antipodal cutoff at affine dimension). *Assume $nd_B > 1$. Then $\alpha_r(\text{Inst}_n(A, B)) = 0$ for all $r \geq D_{A,B}^{(n)}$.*

Proof. Choose affine coordinates $c : \mathfrak{A}_{A,B}^{(n)} \rightarrow \mathbb{R}^D$ (an affine isomorphism, hence injective). For any continuous $h : S^r \rightarrow \text{Inst}_n(A, B)$ with $r \geq D$, restrict to an equatorial $S^D \subset S^r$; Borsuk–Ulam applied to $c \circ h : S^D \rightarrow \mathbb{R}^D$ gives u with $c(h(u)) = c(h(-u))$, hence $h(u) = h(-u)$ by injectivity. Thus $\alpha_D = 0$, and monotonicity (Proposition 4.3) extends this to all $r \geq D$. $\square$

Theorem 4.10 (Observable antipodal sphere). *Assume either (i) $n \geq 2$ and $d_A \geq 2$, or (ii) $n = 1$, $d_A \geq 2$, $d_B \geq 2$. Then*

$$\alpha_r(\text{Inst}_n(A, B)) = 1 \quad (0 \leq r \leq d_A^2 - 2).$$

Proof. First suppose $n \geq 2$. Fix a unit vector $|b_0\rangle \in B$ and let $\sigma = |b_0\rangle\langle b_0|$. Write $\text{Herm}_0(A)$ for the real vector space of traceless Hermitian operators on A , of real dimension $d_A^2 - 1$. For $H \in \text{Herm}_0(A)$, $H \neq 0$, define

$$K(H) = \frac{H}{\|H\|_\infty}, \quad E_\pm(H) = \frac{I_A \pm K(H)}{2}.$$

Then $E_\pm(H) \geq 0$ and $E_+(H) + E_-(H) = I_A$. Define an instrument by

$$\mathcal{I}_1^H(X) = \text{Tr}(E_+(H)X)\sigma, \quad \mathcal{I}_2^H(X) = \text{Tr}(E_-(H)X)\sigma, \quad \mathcal{I}_k^H(X) = 0 \quad (k \geq 3);$$

the sum is the replacement channel $X \mapsto \text{Tr}(X)\sigma$, so $\mathcal{I}^H \in \text{Inst}_n(A, B)$. Since $\|K(H)\|_\infty = 1$, $K(H)$ has a unit eigenvector $|\psi\rangle$ with eigenvalue $+1$ or -1 . If $K(H)|\psi\rangle = |\psi\rangle$, then $\text{Tr}(E_+(H)|\psi\rangle\langle\psi|) = 1$ and $\text{Tr}(E_-(H)|\psi\rangle\langle\psi|) = 0$; for $-H$ the roles of E_+ and E_- are exchanged. Hence the flagged outputs of $\mathcal{I}^H$ and $\mathcal{I}^{-H}$ on the input $|\psi\rangle\langle\psi|$ are $\sigma \otimes |1\rangle\langle 1|$ and $\sigma \otimes |2\rangle\langle 2|$ respectively, which are orthogonal, so their trace norm difference is 2; by the diameter bound the instrument distance is exactly 2. The map $H \mapsto H/\|H\|_\infty$ is continuous on the Euclidean unit sphere of $\text{Herm}_0(A) \cong \mathbb{R}^{d_A^2-1}$, so the construction gives a continuous image of $S^{d_A^2-2}$ with antipodal distance 2; equatorial restriction gives the result for all $0 \leq r \leq d_A^2 - 2$.

For $n = 1$ and $d_B \geq 2$, choose two orthogonal output states σ_+, σ_- and define the channel

$$\Phi^H(X) = \text{Tr}(E_+(H)X)\sigma_+ + \text{Tr}(E_-(H)X)\sigma_-.$$

The same eigenvector witness gives orthogonal outputs σ_+ and σ_- , hence distance 2, and the continuity argument is identical. $\square$

Proposition 4.11 (Orthogonal join of instrument spheres). *Let $W_1, \dots, W_\ell$ be pairwise orthogonal flag-block-diagonal subspaces of $B \otimes C_n$, and suppose that continuous maps $h_i : S^{r_i} \rightarrow \mathcal{S}_{\text{flag}}(B \otimes C_n)$ take values in states supported on W_i and satisfy*

$$\|h_i(u) - h_i(-u)\|_1 = 2 \quad (u \in S^{r_i}).$$

Then there is a continuous map $H : S^{\sum_i r_i + \ell - 1} \rightarrow \mathcal{S}_{\text{flag}}(B \otimes C_n)$ with

$$\|H(y) - H(-y)\|_1 = 2$$

for every y . Composing with the replacement-instrument embedding gives the same antipodal separation in instrument diamond norm.

Proof. Regard the domain sphere as the unit sphere in $\bigoplus_{i=1}^\ell \mathbb{R}^{r_i+1}$. For $y = (y_1, \dots, y_\ell)$ set

$$\lambda_i = \|y_i\|_2^2, \quad u_i = \frac{y_i}{\|y_i\|_2} \quad \text{when } y_i \neq 0,$$

and define

$$H(y) = \sum_{i: y_i \neq 0} \lambda_i h_i(u_i),$$

with the i -th summand set to zero when $y_i = 0$. Since $\sum_i \lambda_i = 1$, $H(y)$ is a flagged state. Continuity at points where some $y_i = 0$ follows because the trace norm of the i -th summand is $\|\lambda_i h_i(u_i)\|_1 = \lambda_i \rightarrow 0$ as $y_i \rightarrow 0$, uniformly in the direction u_i . Because the supports W_i are pairwise orthogonal, the difference $H(y) - H(-y)$ is block diagonal with respect to $\bigoplus_i W_i$, so

$$\|H(y) - H(-y)\|_1 = \sum_i \lambda_i \|h_i(u_i) - h_i(-u_i)\|_1 = 2 \sum_i \lambda_i = 2.$$

Lemma 4.4 transfers the equality to replacement instruments. $\square$

Proposition 4.12 (Instrument-level join with replacement spheres). *Let $W \subseteq B \otimes C_n$ be a flag-block-diagonal subspace. Let $h : S^{r_1} \rightarrow \text{Inst}_n(A, B)$ be continuous and such that*

1. *for every input state ρ on $A_0 \otimes A$ and every $u \in S^{r_1}$, the flagged output $(\text{id}_{A_0} \otimes \widehat{h(u)})(\rho)$ is supported on $A_0 \otimes W$, and*
2. *$\|h(u) - h(-u)\|_{\diamond, \text{inst}} = 2$ for every $u \in S^{r_1}$.*

Let $\tau : S^{r_2} \rightarrow \mathcal{S}_{\text{flag}}(B \otimes C_n)$ be continuous, take values in states supported on the per-block orthogonal complement $W^\perp$, and satisfy $\|\tau(v) - \tau(-v)\|_1 = 2$ for every $v \in S^{r_2}$. Then, on the unit sphere of $\mathbb{R}^{r_1+1} \oplus \mathbb{R}^{r_2+1}$,

$$\mathcal{I}(y) = \|y_1\|_2^2 h\left(\frac{y_1}{\|y_1\|_2}\right) + \|y_2\|_2^2 \vec{\mathcal{R}}_{\tau(y_2/\|y_2\|_2)}$$

(with a summand read as zero when the corresponding coordinate block vanishes) defines a continuous map $S^{r_1+r_2+1} \rightarrow \text{Inst}_n(A, B)$ with

$$\|\mathcal{I}(y) - \mathcal{I}(-y)\|_{\diamond, \text{inst}} = 2 \quad (y \in S^{r_1+r_2+1}).$$

Proof. Both summands are convex combinations of instruments, hence instruments, so $\mathcal{I}(y) \in \text{Inst}_n(A, B)$; the instrument body is bounded (each flagged channel has diamond norm 1), so each summand tends to zero in norm as its weight tends to zero, uniformly over the sphere, which gives continuity on the coordinate spheres $\{y_i = 0\}$ exactly as in Proposition 4.11. Fix y and write $\lambda = \|y_1\|_2^2$, $u = y_1/\|y_1\|_2$, $v = y_2/\|y_2\|_2$ where the corresponding blocks are nonzero. If $\lambda = 0$ the claim reduces to the replacement pair, at distance 2 by Lemma 4.4. Otherwise set $\vec{\Delta}_1 = h(u) - h(-u)$ and $\vec{\Delta}_2 = \vec{\mathcal{R}}_{\tau(v)} - \vec{\mathcal{R}}_{\tau(-v)}$, so that $\mathcal{I}(y) - \mathcal{I}(-y) = \lambda \vec{\Delta}_1 + (1 - \lambda) \vec{\Delta}_2$. The set of input states is compact and $\rho \mapsto \|(\text{id} \otimes \vec{\Delta}_1)(\rho)\|_1$ is continuous with supremum $\|\vec{\Delta}_1\|_{\diamond, \text{inst}} = 2$, so there is an input state ρ_* with $\|(\text{id} \otimes \vec{\Delta}_1)(\rho_*)\|_1 = 2$. For this same input, Lemma 4.4 gives $(\text{id} \otimes \vec{\Delta}_2)(\rho_*) = \text{Tr}_A(\rho_*) \otimes (\tau(v) - \tau(-v))$, of trace norm $\|\tau(v) - \tau(-v)\|_1 = 2$, independently of ρ_* . By (i) applied to $h(u)$ and $h(-u)$, the Hermitian operator $(\text{id} \otimes \vec{\Delta}_1)(\rho_*)$ is supported on $A_0 \otimes W$, while $(\text{id} \otimes \vec{\Delta}_2)(\rho_*)$ is supported on $A_0 \otimes W^\perp$; orthogonal supports make the trace norm additive, so

$$\|(\text{id} \otimes (\mathcal{I}(y) - \mathcal{I}(-y)))(\rho_*)\|_1 = \lambda \cdot 2 + (1 - \lambda) \cdot 2 = 2.$$

Hence $\|\mathcal{I}(y) - \mathcal{I}(-y)\|_{\diamond, \text{inst}} \geq 2$, and the reverse inequality is the diameter bound (Proposition 3.9). $\square$

Remark 4.13 (Why the join carries at most one input-dependent sphere). *Proposition 4.12 joins one input-dependent sphere with arbitrarily many replacement spheres: the replacement components attain their maximal separation 2 on every input, so the eigenvector input that exposes the input-dependent component simultaneously exposes all of them. Two input-dependent components could not be joined by this mechanism, since their separating inputs (eigenvectors of the respective observables) vary with the sphere parameter and need not coincide; this is why exactly one observable sphere enters Corollary 4.14 below.*

Corollary 4.14 (Observable-replacement join dimension). *Under the hypotheses of Theorem 4.10, the flagged outputs of the observable sphere are supported on the flagged subspace*

$$W_{\text{obs}} = \text{span}\{|b_0\rangle \otimes |1\rangle, |b_0\rangle \otimes |2\rangle\}$$

in the case $n \geq 2$ (with the pure state $\sigma = |b_0\rangle\langle b_0|$), and on the 2-dimensional flagged subspace $W_{\text{obs}} = \text{span}\{\sigma_+, \sigma_-\}$ in the case $n = 1$. Let $W_{\text{obs}}^\perp$ denote the per-block orthogonal complement of W_{obs} : its first two blocks are $\{|b_0\rangle\}^\perp \subseteq B$ and its remaining $n - 2$ blocks are all of B when

$n \geq 2$, and it is the orthogonal complement of W_{obs} in B when $n = 1$. The real vector space of flag-block-diagonal Hermitian operators supported on $W_{\text{obs}}^\perp$ has dimension

$$N_{\text{join}} = \begin{cases} 2(d_B - 1)^2 + (n - 2)d_B^2, & n \geq 2, \\ (d_B - 2)^2, & n = 1. \end{cases}$$

If $N_{\text{join}} \geq 2$, then

$$\alpha_r(\text{Inst}_n(A, B)) = 1 \quad (0 \leq r \leq d_A^2 + N_{\text{join}} - 3).$$

Proof. The observable sphere $h : S^{d_A^2-2} \rightarrow \text{Inst}_n(A, B)$ of Theorem 4.10 is continuous and has antipodal instrument-diamond distance 2; on every input ρ , each outcome component of its flagged output is a positive operator on A_0 tensored with $\sigma \otimes |k\rangle\langle k|$, $k \in \{1, 2\}$ (respectively with σ_+ or σ_- when $n = 1$), and the remaining outcomes vanish, so all flagged outputs are supported on $A_0 \otimes W_{\text{obs}}$. The per-block complement $W_{\text{obs}}^\perp$ is itself flag-block diagonal, and the Jordan construction of Theorem 4.5, applied to the traceless part of the real vector space of flag-block-diagonal Hermitian operators supported on $W_{\text{obs}}^\perp$, produces a continuous sphere $\tau : S^{N_{\text{join}}-2} \rightarrow \mathcal{S}_{\text{flag}}(B \otimes C_n)$ of states supported on $W_{\text{obs}}^\perp$ with $\|\tau(v) - \tau(-v)\|_1 = 2$; the condition $N_{\text{join}} \geq 2$ is exactly nonemptiness of this sphere. Proposition 4.12 with $W = W_{\text{obs}}$ joins the two spheres into a continuous map

$$S^{(d_A^2-2)+(N_{\text{join}}-2)+1} = S^{d_A^2+N_{\text{join}}-3} \longrightarrow \text{Inst}_n(A, B)$$

with antipodal instrument-diamond distance 2; equatorial restriction supplies every smaller index, and $\alpha_r \leq 1$ always by the diameter bound (Proposition 3.9). $\square$

Remark 4.15 (Structured partitions). *The ℓ -fold join of Proposition 4.11 may also be applied to arbitrary pairwise orthogonal output-flag blocks (B_i, n_i) with $\sum_i b_i \leq d_B$, $\sum_i n_i = n$, producing a sphere of dimension $\sum_i (n_i b_i^2 - 1) - 1$ whenever $n_i b_i^2 \geq 2$ for every retained block. Taking one block with $b_1 = d_B$, $n_1 = n$ recovers the leading dimension $nd_B^2 - 2$; no partition improves Theorem 4.5.*

Definition 4.16 (Certified one-sphere dimension). *Let*

$$L_{\text{rep}} = nd_B^2 - 2,$$

valid whenever $nd_B > 1$. If either $n \geq 2, d_A \geq 2$, or $n = 1, d_A \geq 2, d_B \geq 2$, set $L_{\text{obs}} = d_A^2 - 2$; otherwise set $L_{\text{obs}} = -1$. If the observable sphere is valid and $N_{\text{join}} \geq 2$ (Corollary 4.14), set $L_{\text{join}} = d_A^2 + N_{\text{join}} - 3$; otherwise $L_{\text{join}} = -1$. Finally define

$$L_{A,B}^{(n)} = \min \left\{ D_{A,B}^{(n)} - 1, \max \{ L_{\text{rep}}, L_{\text{obs}}, L_{\text{join}} \} \right\},$$

where invalid terms (-1) are ignored. If $d_A = 1$, the observable and join terms are invalid and $L_{1,B}^{(n)} = nd_B^2 - 2 = D_{1,B}^{(n)} - 1$.

Definition 4.17 (Strengthened certified lower envelope). *Assume $nd_B > 1$. Let*

$$\rho_n = \frac{2}{nd_B \min\{d_A, d_B\}}, \quad \gamma_{A,B}^{(n)} = \max \left\{ \rho_n, \frac{1}{d_A} \right\}.$$

Define

$$\beta_r^{\text{best},(n)}(A, B) = \begin{cases} 1, & 0 \leq r \leq L_{A,B}^{(n)}, \\ \gamma_{A,B}^{(n)}, & L_{A,B}^{(n)} < r < D_{A,B}^{(n)}, \\ 0, & r \geq D_{A,B}^{(n)}. \end{cases}$$

When $L_{A,B}^{(n)} = D_{A,B}^{(n)} - 1$ the middle range is empty (this is always the case for $d_A = 1$). Every component of the envelope is rigorously certified: the value 1 on the first range by Theorems 4.5 and 4.10 and Corollary 4.14, the value $\gamma_{A,B}^{(n)}$ on the middle range by Theorems 3.3 and 4.7, and the value 0 on the last range by Theorem 5.7. Within the ball mechanism the term ρ_n is optimal over all centres of the body (Theorem 3.12).

Remark 4.18 (Resource interpretation of the two breakpoints). *The two breakpoints of the envelope have a direct resource interpretation. A flagged output state has $nd_B^2 - 1$ affine parameters, while an input-dependent instrument has $d_A^2(nd_B^2 - 1)$. The first regime ($r \leq L_{A,B}^{(n)}$) is already obstructed by preparing and recording one classical–quantum output state — augmented by the observable constructions, which obstruct input-dependent effect readout even when the output state is fixed. The second regime concerns the additional dependence of all outcome branches on the input operator. Increasing the number of outcomes therefore has two opposite effects: it linearly enlarges the global representation space, while dividing the local positivity margin of the uniform instrument by n .*

5 Compression widths: upper bounds and exact thresholds

Convention 5.1 (Continuous encoding, unrestricted decoding). *Throughout the compression-width results, an encoder is required to be continuous while the decoder is unrestricted unless explicitly stated otherwise. This convention is essential for the Helly-type upper bounds; where a continuous decoder is available we state this separately.*

Definition 5.2 (Euclidean instrument width).

$$\delta_{r,\text{inst}}^\diamond(A, B) = \inf_{\substack{f: \text{Inst}_n(A, B) \rightarrow \mathbb{R}^r \\ g: \mathbb{R}^r \rightarrow \text{Inst}_n(A, B)}} \sup_{\text{continuous } \mathcal{I} \in \text{Inst}_n(A, B)} \|g(f(\mathcal{I})) - \mathcal{I}\|_{\diamond, \text{inst}}.$$

For compact metric $K \subseteq L$ we also use $\delta_r^c(K; L)$, the analogous width with arbitrary decoder into L .

Remark 5.3 (Decoder codomain does not weaken the lower bounds). *Since $\text{RepInst}_n(A, B) \subset \text{Inst}_n(A, B) \subset \mathfrak{A}_{A,B}^{(n)}$, and the triangle inequality $\|h(u) - \Psi\|_{\diamond, \text{inst}} + \|h(-u) - \Psi\|_{\diamond, \text{inst}} \geq \|h(u) - h(-u)\|_{\diamond, \text{inst}}$ holds for every Ψ in the ambient vector space $\bigoplus_k L(A, B)$, the lower bounds of Corollaries 5.5 and 5.6 are proved for decoders with values in the full affine space $\mathfrak{A}_{A,B}^{(n)}$ — a strictly stronger statement than for instrument-valued decoders. Hence*

$$\beta_r^{\text{best}, (n)}(A, B) \leq \delta_r^c(\text{Inst}_n(A, B); \mathfrak{A}_{A,B}^{(n)}) \leq \delta_r^c(\text{Inst}_n(A, B); \text{Inst}_n(A, B)) = \delta_{r,\text{inst}}^\diamond(A, B),$$

$$\delta_{r,\text{inst}}^\diamond(A, B) \leq U_r^{(n)}(A, B).$$

Allowing unphysical affine decoders cannot circumvent the certified lower envelope; and because the optimal constant-code centre $\mathcal{I}_$ is itself an instrument (Theorem 3.8), the constant-code width coincides over both codomains (Corollary 5.14), so every upper bound of this paper is admissible with a physical instrument decoder.*

Corollary 5.4 (Certified width bracket). *Assume $nd_B > 1$. For $0 \leq r < D_{A,B}^{(n)}$,*

$$\beta_r^{\text{best}, (n)}(A, B) \leq \delta_{r,\text{inst}}^\diamond(A, B) \leq U_r^{(n)}(A, B),$$

where $U_r^{(n)}$ is defined in Definition 5.12. For $r \geq D_{A,B}^{(n)}$ the width is zero.

Proof. For $0 \leq r \leq L_{A,B}^{(n)}$ the replacement sphere, observable sphere, or observable–replacement join supplies a continuous map $h : S^r \rightarrow \text{Inst}_n(A, B)$ with antipodal distance 2; composing with any continuous encoder f gives $f \circ h : S^r \rightarrow \mathbb{R}^r$, and Borsuk–Ulam produces u with $f(h(u)) = f(h(-u))$. The decoder returns one reconstruction for the pair, and the triangle inequality forces error at least $1 = \beta_r^{\text{best},(n)}$. The argument applies verbatim to decoders valued in the ambient affine space $\mathfrak{A}_{A,B}^{(n)}$ (Remark 5.3), which only strengthens the lower bound. For $L_{A,B}^{(n)} < r < D_{A,B}^{(n)}$, the centred ball gives antipodal separation $2\rho_n$ (via the radial map of Corollary 5.5, which needs only $r + 1 \leq D_{A,B}^{(n)}$) while Theorem 4.7 gives antipodal separation $2/d_A$; applying the Borsuk–Ulam argument to the two spheres separately forces error at least ρ_n and at least $1/d_A$, hence at least $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$. The upper terms are Theorems 3.8 and 5.11 and Proposition 5.8; exact reconstruction for $r \geq D_{A,B}^{(n)}$ is Theorem 5.7. $\square$

For a Hausdorff latent space Y , recall the deleted product

$$F_2(Y) = \{(y_0, y_1) \in Y^2 : y_0 \neq y_1\}$$

with the involution exchanging the two entries, and let

$$\text{coind}_{\mathbb{Z}_2} F_2(Y) = \sup \{k \geq 0 : \exists \text{ equivariant } S^k \rightarrow F_2(Y)\},$$

with the convention $\text{coind}_{\mathbb{Z}_2} \emptyset = -1$ and the value $+\infty$ also allowed. The deleted product carries the subspace topology of Y^2 and the free $\mathbb{Z}_2$ -action exchanging the entries; since Y is Hausdorff, the diagonal of Y^2 is closed and $F_2(Y)$ is its open complement. Only the existence of equivariant maps into $F_2(Y)$ is used below. If $\text{coind}_{\mathbb{Z}_2} F_2(Y) = +\infty$, no integer q satisfies $q > \text{coind}_{\mathbb{Z}_2} F_2(Y)$ and the corollary below is vacuous. If Y admits a continuous injection $j : Y \rightarrow \mathbb{R}^r$, the normalised difference map

$$(y_0, y_1) \mapsto \frac{j(y_0) - j(y_1)}{\|j(y_0) - j(y_1)\|_2}$$

is continuous and equivariant $F_2(Y) \rightarrow S^{r-1}$, so by Borsuk–Ulam [9], $\text{coind}_{\mathbb{Z}_2} F_2(Y) \leq r - 1$.

Corollary 5.5 (Intrinsic latent-space instrument hierarchy). *Assume $nd_B > 1$. Let Y be Hausdorff and let $q \in \mathbb{N}_0$ satisfy*

$$q > \text{coind}_{\mathbb{Z}_2} F_2(Y).$$

Then every continuous encoder $\text{Enc} : \text{Inst}_n(A, B) \rightarrow Y$ and every decoder $\text{Dec} : Y \rightarrow \mathfrak{A}_{A,B}^{(n)}$ satisfy

$$\sup_{\mathcal{I} \in \text{Inst}_n(A, B)} \|\text{Dec}(\text{Enc}(\mathcal{I})) - \mathcal{I}\|_{\diamond, \text{inst}} \geq \beta_q^{\text{best},(n)}(A, B).$$

Proof. There is nothing to prove when $q \geq D_{A,B}^{(n)}$. If $0 \leq q \leq L_{A,B}^{(n)}$, restrict the perfectly separated sphere supplied by Theorem 4.5, Theorem 4.10, or Corollary 4.14 to an equatorial S^q . If $L_{A,B}^{(n)} < q < D_{A,B}^{(n)}$, then $q + 1 \leq D_{A,B}^{(n)}$; choose a $(q + 1)$ -dimensional subspace $W \subseteq \mathfrak{V}_{A,B}^{(n)}$ and use the radial map

$$h(u) = \mathcal{I}_* + \rho_n \frac{u}{\|u\|_{\diamond, \text{inst}}}$$

on an auxiliary Euclidean sphere $S^q \subset W$ (the denominator is nonzero on the Euclidean unit sphere, and finite-dimensional norm equivalence makes the normalisation continuous). Theorem 3.3 makes this map instrument-valued, and its antipodal separation is $2\rho_n$; alternatively, the full-direction sphere of Theorem 4.7, restricted to an equatorial S^q , has antipodal separation $2/d_A$. In both regimes, if the two latent codes $\text{Enc}(h(u))$ and $\text{Enc}(h(-u))$ were distinct for every u , then $u \mapsto (\text{Enc}(h(u)), \text{Enc}(h(-u)))$ would be an equivariant map $S^q \rightarrow F_2(Y)$, contradicting the coindex hypothesis. An antipodal pair therefore has one code; the decoder returns one reconstruction for it, and the triangle inequality gives half the antipodal separation of the sphere

used. In the first regime the half-separation is $1 = \beta_q^{\text{best},(n)}(A, B)$. In the second regime the radial sphere and the equatorial restriction of the full-direction sphere, used separately, give error at least ρ_n and at least $1/d_A$, hence at least $\beta_q^{\text{best},(n)}(A, B) = \gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$. $\square$

Corollary 5.6 (Continuous instrument-compression hierarchy). *Assume $nd_B > 1$, and let Y admit a continuous injection into $\mathbb{R}^r$. Then, for every continuous encoder $\text{Enc} : \text{Inst}_n(A, B) \rightarrow Y$ and every decoder $\text{Dec} : Y \rightarrow \mathfrak{A}_{A,B}^{(n)}$,*

$$\sup_{\mathcal{I} \in \text{Inst}_n(A, B)} \|\text{Dec}(\text{Enc}(\mathcal{I})) - \mathcal{I}\|_{\diamond, \text{inst}} \geq \beta_r^{\text{best},(n)}(A, B).$$

Proof. If $r = 0$ then Y has at most one point and $F_2(Y) = \emptyset$ has coindex -1 ; if $r \geq 1$ the normalised difference map gives $\text{coind}_{\mathbb{Z}_2} F_2(Y) \leq r - 1$ as above. Apply Corollary 5.5 with $q = r$. $\square$

Theorem 5.7 (Zero-error threshold, with continuous decoding at the threshold). *For Euclidean latent spaces and under Convention 5.1,*

$$\delta_{r, \text{inst}}^{\diamond}(A, B) = 0 \iff r \geq D_{A,B}^{(n)}.$$

Moreover, for every $r \geq D_{A,B}^{(n)}$, exact decoding may be chosen continuous.

Proof. For $r \geq D_{A,B}^{(n)}$, choose affine coordinates $c : \mathfrak{A}_{A,B}^{(n)} \rightarrow \mathbb{R}^D$ (an affine isomorphism) and restrict to the compact convex body $K = \text{Inst}_n(A, B)$; its image $C = c(K) \subset \mathbb{R}^D$ is compact and convex. The encoder is $f = c|_K$, padded with zeros if $r > D$. Let $\Pi_C : \mathbb{R}^D \rightarrow C$ be the Euclidean nearest-point projection: for a nonempty closed convex subset of Euclidean space the metric projection is single-valued and 1-Lipschitz, hence continuous. Define the decoder $g(z) = c^{-1}(\Pi_C(z_1, \dots, z_D))$, ignoring any padding. Then $g(f(x)) = x$ for every $x \in K$ and g is continuous. For the converse, if $r < D_{A,B}^{(n)}$ then $\beta_r^{\text{best},(n)}(A, B) > 0$ (Corollary 5.6), so exact reconstruction is impossible even with an arbitrary decoder; the collision $f(h(u)) = f(h(-u))$ forces one decoder value to cover two distinct instruments, using only that the *code values coincide*, with no regularity assumption on the decoder. $\square$

Proposition 5.8 (Convex-projection instrument bound). *For $0 \leq r \leq D_{A,B}^{(n)}$,*

$$\delta_{r, \text{inst}}^{\diamond}(A, B) \leq \frac{2(D_{A,B}^{(n)} - r)}{D_{A,B}^{(n)} - r + 1}.$$

Proof. For $r = D_{A,B}^{(n)}$ affine coordinates give exact reconstruction. Otherwise project affine coordinates of the compact convex instrument body onto $\mathbb{R}^r$ and put $k = D_{A,B}^{(n)} - r$. Each nonempty fibre F is compact and convex, has diameter at most two (Proposition 3.9), and has affine dimension at most k . Set $R = 2k/(k+1)$ and consider the family $\{F\} \cup \{\overline{B}(x, R) : x \in F\}$ in $\text{aff}(F) \simeq \mathbb{R}^k$. Any subfamily of at most $k+1$ members intersects: if it contains m balls centred at $x_1, \dots, x_m$, their barycentre $c = \frac{1}{m} \sum_j x_j$ lies in F by convexity and satisfies

$$\|c - x_j\| \leq \frac{2(m-1)}{m} \leq \frac{2k}{k+1} = R$$

(the second inequality is equivalent to $m \leq k+1$); if F is also a member of the subfamily then $m \leq k$. Helly's theorem [10] and compactness of F therefore give a point of F within distance R of the entire fibre. Use these points as decoder values. $\square$

Remark 5.9 (Nonconstructive Helly decoder). *The decoder in Proposition 5.8 is selected via Helly's theorem and compactness for each fibre; it is guaranteed to take values in the fibre and to achieve error $\leq 2k/(k+1)$, but its continuity is not claimed and is not required by Definition 5.2. The word “certified” refers to proved upper bounds, not to continuous or algorithmic decoders. If both encoder and decoder were required to be continuous, the zero-error threshold $r \geq D_{A,B}^{(n)}$ remains valid (Theorem 5.7 supplies a continuous decoder), while the subcritical upper bounds become existence statements whose Helly-selected decoders may fail continuity.*

Lemma 5.10 (Exact pinching norm). *For an orthogonal decomposition $H = \bigoplus_{j=1}^{\ell} H_j$ into nonzero blocks and the channel $\mathcal{P}_H(X) = \sum_j P_j X P_j$,*

$$\|\text{id}_H - \mathcal{P}_H\|_{\diamond} = 2\left(1 - \frac{1}{\ell}\right).$$

Proof. Set $\omega = e^{2\pi i/\ell}$ and $U_m = \sum_j \omega^{m(j-1)} P_j$. The roots-of-unity identity $\frac{1}{\ell} \sum_{m=0}^{\ell-1} \omega^{m(j-k)} = \delta_{jk}$ gives $\mathcal{P}_H = \frac{1}{\ell} \sum_{m=0}^{\ell-1} \mathcal{U}_{U_m}$, so

$$\|\text{id}_H - \mathcal{P}_H\|_{\diamond} \leq \frac{1}{\ell} \sum_{m=1}^{\ell-1} \|\text{id}_H - \mathcal{U}_{U_m}\|_{\diamond} \leq 2\left(1 - \frac{1}{\ell}\right).$$

For equality choose unit vectors $|e_j\rangle \in H_j$ and apply the difference map to the normalised test vector $|e\rangle = \ell^{-1/2} \sum_j |e_j\rangle$; on their span the output is represented by $(J_{\ell} - I_{\ell})/\ell$, whose trace norm is $2(1 - 1/\ell)$. $\square$

Theorem 5.11 (Input- and output-pinching for instruments). *The following bounds hold.*

1. *If $B = \bigoplus_{j=1}^{\ell_{\text{out}}} B_j$, $b_j = \dim B_j > 0$, and*

$$r_{\text{out}}^{(n)} = d_A^2 \left(n \sum_j b_j^2 - 1 \right),$$

then, for $r \geq r_{\text{out}}^{(n)}$,

$$\delta_{r,\text{inst}}^{\diamond}(A, B) \leq 2\left(1 - \frac{1}{\ell_{\text{out}}}\right).$$

2. *If $A = \bigoplus_{i=1}^{k_{\text{in}}} A_i$, $a_i = \dim A_i > 0$, and*

$$r_{\text{in}}^{(n)} = \left(\sum_i a_i^2 \right) (n d_B^2 - 1),$$

then, for $r \geq r_{\text{in}}^{(n)}$,

$$\delta_{r,\text{inst}}^{\diamond}(A, B) \leq 2\left(1 - \frac{1}{k_{\text{in}}}\right).$$

Proof. For output pinching, the flagged Choi operators have one Hermitian block on $A_0 \otimes B_j$ for every outcome and output block, of ambient real dimension $n d_A^2 \sum_j b_j^2$. The summed marginal map onto $\text{Herm}(A_0)$ is surjective: for prescribed $H \in \text{Herm}(A_0)$ place $H \otimes I_{B_1}/b_1$ in one outcome block and set the others to zero, since $\text{Tr}_{B_1}(H \otimes I_{B_1}/b_1) = H$. Thus the marginal rank is d_A^2 , and the uniform depolarising instrument is positive definite in the relative affine space, so the output-pinched body has affine dimension $r_{\text{out}}^{(n)}$. Encode the output-pinched instrument by affine coordinates and invert on the image; the reconstruction of $\mathcal{I}$ is $(\mathcal{P}_B \otimes \text{id}_{C_n}) \circ \mathcal{I}$, and by composition submultiplicativity of the diamond norm [6] and Lemma 5.10,

$$\|\mathcal{I} - (\mathcal{P}_B \otimes \text{id}_{C_n}) \circ \mathcal{I}\|_{\diamond, \text{inst}} \leq \|\text{id}_B - \mathcal{P}_B\|_{\diamond} \cdot \|\widehat{\mathcal{I}}\|_{\diamond} = 2\left(1 - \frac{1}{\ell_{\text{out}}}\right).$$

For input pinching, use the basis-conjugate decomposition $A_0 = \bigoplus_i \bar{A}_i$: with the fixed Choi basis, the input pinching channel $\mathcal{P}_A = \sum_i P_i(\cdot)P_i$ induces on Choi operators

$$\hat{\mathcal{J}}_{\Phi \circ \mathcal{P}_A} = (\mathcal{P}_{\bar{A}_0} \otimes \text{id}_B) \hat{\mathcal{J}}_\Phi,$$

where $\bar{A}_i$ is the conjugate subspace in A_0 of the block $A_i \subseteq A$. The block-diagonal flagged Choi space has dimension $nd_B^2 \sum_i a_i^2$, and the blockwise marginal map is surjective: for prescribed $H_i \in \text{Herm}(\bar{A}_i)$ place $H_i \otimes I_B/d_B$ in one outcome block. Thus the constraints have total rank $\sum_i a_i^2$, and the same positive-definite witness gives affine dimension $r_{\text{in}}^{(n)}$. Encoding $\mathcal{I} \circ \mathcal{P}_A$ by affine coordinates and inverting on the image gives

$$\|\mathcal{I} - \mathcal{I} \circ \mathcal{P}_A\|_{\diamond, \text{inst}} \leq \|\hat{\mathcal{I}}\|_{\diamond} \cdot \|\text{id}_A - \mathcal{P}_A\|_{\diamond} = 2\left(1 - \frac{1}{k_{\text{in}}}\right).$$

Padding by zero coordinates handles larger r in both cases. $\square$

For $1 \leq \ell \leq d$ write $d = m\ell + t$, $0 \leq t < \ell$, and set

$$Q_\ell(d) = (\ell - t)m^2 + t(m + 1)^2.$$

If two block sizes satisfy $a \geq b + 2$, replacing them by $a - 1$ and $b + 1$ changes their squared sum by $-2(a - b - 1) < 0$; since each exchange strictly decreases the sum and there are only finitely many partitions of d into ℓ positive parts, iteration must terminate at a partition with no two blocks differing by more than 1, i.e. the balanced partition with $\ell - t$ blocks of size m and t blocks of size $m + 1$, where $d = m\ell + t$, $0 \leq t < \ell$. Hence balanced blocks minimise $\sum_j b_j^2$ at fixed ℓ , and $Q_\ell(d) = (\ell - t)m^2 + t(m + 1)^2$.

Definition 5.12 (Certified instrument upper envelope). *For $0 \leq r \leq D_{A,B}^{(n)}$, let $U_r^{(n)}(A, B)$ be the minimum of*

$$2\left(1 - \frac{1}{nd_B \min\{d_A, d_B\}}\right), \quad \frac{2(D_{A,B}^{(n)} - r)}{D_{A,B}^{(n)} - r + 1},$$

all output-pinching values $2(1 - 1/\ell_{\text{out}})$ satisfying $r \geq d_A^2(nQ_{\ell_{\text{out}}}(d_B) - 1)$, and all input-pinching values $2(1 - 1/k_{\text{in}})$ satisfying $r \geq Q_{k_{\text{in}}}(d_A)(nd_B^2 - 1)$. An empty pinching family is ignored.

Proposition 5.13 (Error-one floor of the certified upper envelope). *For $nd_B > 1$ and $0 \leq r < D_{A,B}^{(n)}$, every admissible upper-bound term is at least 1, hence $U_r^{(n)}(A, B) \geq 1$.*

Proof. The constant-code bound satisfies $R_n = 2(1 - 1/(nd_B s)) \geq 1$ because $nd_B s \geq 2$. The convex-projection bound satisfies $2(D - r)/(D - r + 1) \geq 1$ because $D - r \geq 1$, with equality only at $r = D - 1$. Output pinching gives $2(1 - 1/\ell_{\text{out}}) \geq 1$ whenever $\ell_{\text{out}} \geq 2$; $\ell_{\text{out}} = 1$ requires $r \geq d_A^2(nQ_1(d_B) - 1) = d_A^2(nd_B^2 - 1) = D$. Similarly $k_{\text{in}} \geq 2$ below D , because $Q_1(d_A)(nd_B^2 - 1) = D$. $\square$

Corollary 5.14 (Exact constant-code instrument width). *Assume $nd_B > 1$. Then*

$$\delta_{0, \text{inst}}^\diamond(A, B) = R_{\text{inst}}(\mathcal{I}_*) = 2\left(1 - \frac{1}{nd_B s}\right).$$

Proof. The lower bound in Theorem 3.8 holds even for centres Ψ in the full affine space $\mathfrak{A}_{A,B}^{(n)}$, and the optimal centre $\mathcal{I}_*$ is itself an instrument, so the infimum over instruments equals the covering radius. $\square$

Corollary 5.15 (Exact top-subcritical replacement-instrument width). *Assume $nd_B > 1$. Then*

$$\delta_{nd_B^2 - 2}^c(\text{RepInst}_n(A, B); \text{RepInst}_n(A, B)) = 1.$$

When $d_A = 1$ the same equality holds for the full instrument body.

Proof. Compose any continuous encoder into $\mathbb{R}^{nd_B^2-2}$ with the perfectly separated sphere of Theorem 4.5; Borsuk–Ulam and the triangle inequality give the lower bound 1. The replacement body is compact, convex, of affine dimension $m = nd_B^2 - 1$ and diameter two; at the top subcritical index $r = m - 1$ the convex-projection bound of Proposition 5.8 gives

$$\frac{2(m-r)}{m-r+1} = \frac{2}{2} = 1,$$

so the upper bound matches. For $d_A = 1$ every instrument is a replacement instrument. $\square$

Theorem 5.16 (Exact collapse when $nd_Bs = 2$). *Assume $nd_B > 1$ and $nd_B \min\{d_A, d_B\} = 2$. Then*

$$\delta_{r,\text{inst}}^\diamond(A, B) = \begin{cases} 1, & 0 \leq r < D_{A,B}^{(n)}, \\ 0, & r \geq D_{A,B}^{(n)}. \end{cases}$$

This occurs for (i) arbitrary-input two-outcome POVMs ($d_B = 1$, $n = 2$, any d_A), where $D_{A,1}^{(2)} = d_A^2$, and (ii) trivial-input qubit channels ($d_A = 1$, $n = 1$, $d_B = 2$), where $D_{1,2}^{(1)} = 3$. Thus continuous compression of all two-outcome POVMs is solved exactly: error 1 until full affine dimension, then 0.

Proof. If $nd_Bs = 2$, then $\rho_n = 1$ and $R_n = 1$. For every $r < D_{A,B}^{(n)}$, choose a subspace $W \subseteq \mathfrak{V}_{A,B}^{(n)}$ of dimension $r + 1$ and define on the auxiliary Euclidean sphere $S^r \subset W$ the radial map

$$h(u) = \mathcal{I}_* + \rho_n \frac{u}{\|u\|_{\diamond, \text{inst}}}.$$

Because $\rho_n = 1$ and Theorem 3.3 applies, $h(u)$ is an instrument for every u , and $\|h(u) - h(-u)\|_{\diamond, \text{inst}} = 2\rho_n = 2$. By Borsuk–Ulam every continuous encoder to $\mathbb{R}^r$ and every decoder must produce error at least 1. Conversely, the constant-code bound gives $\delta_{0,\text{inst}}^\diamond \leq R_n = 1$, and by monotonicity of the width the value is at most 1 for all r . Hence the width is exactly 1 for $0 \leq r < D_{A,B}^{(n)}$, and 0 at and above $D_{A,B}^{(n)}$ by Theorem 5.7. $\square$

Proposition 5.17 (Gap structure and endpoint exactness). *Let $m_n = nd_B^2 - 1$, $D_n = D_{A,B}^{(n)}$ (Proposition 3.1), $\rho_n = 2/(nd_Bs)$, $R_n = 2 - \rho_n$. Then:*

1. *At $r = 0$ the width is exactly R_n , so the exact gap is $R_n - 1 = 1 - \rho_n$.*
2. *For $0 < r \leq L_{A,B}^{(n)}$, $\beta_r^{\text{best},(n)} = 1$ and $U_r^{(n)} \geq 1$; the true width satisfies $1 \leq \delta_{r,\text{inst}}^\diamond \leq U_r^{(n)}$ and is not determined except in the cases below.*
3. *For $L_{A,B}^{(n)} < r < D_n$, $\beta_r^{\text{best},(n)} = \gamma_{A,B}^{(n)}$ and $U_r^{(n)} \geq 1$, so the certified bracket has length at least $1 - \gamma_{A,B}^{(n)}$; this bounds the certificate, not the unknown true width, and the ball contribution ρ_n to $\gamma_{A,B}^{(n)}$ is optimal over all centres (Theorem 3.12).*
4. *For $r \geq D_n$ the width is exactly 0.*
5. *For the replacement subfamily, $\delta_{m_n-1}^c(\text{RepInst}_n; \text{RepInst}_n) = 1$ exactly, and this equals the full body when $d_A = 1$.*
6. *When $nd_Bs = 2$ the full width collapses to the step function of Theorem 5.16.*

Proof. Item 1: at $r = 0$ no pinching term is admissible below D_n , because a pinching family with at least two blocks has threshold at least $d_A^2(n \cdot 2 - 1) \geq d_A^2 \geq 1$ for output pinching and at least $2(nd_B^2 - 1) \geq 2$ for input pinching, while a single block gives the trivial threshold D_n itself; hence U_0 is the minimum of R_n and $2D_n/(D_n + 1)$, and

$$2 - \frac{2}{D_n + 1} \geq 2 - \frac{2}{nd_Bs} = R_n$$

because $D_n + 1 = d_A^2(nd_B^2 - 1) + 1 \geq nd_B^2 \geq nd_B$; hence $U_0 = R_n$, and with $\beta_0^{\text{best},(n)} = 1$ the gap is exactly $R_n - 1 = 1 - \rho_n$. Items 2–6 restate the bracket (Corollary 5.4), the floor (Proposition 5.13), Corollaries 5.14 and 5.15, and Theorems 5.7 and 5.16; compare also Corollary 4.9 for the antipodal profile at and above the affine dimension. $\square$

Corollary 5.18 (Asymptotic positivity–dimension product).

$$\begin{aligned}\Pi_n &:= D_{A,B}^{(n)} \rho_{\diamond, \text{inst}}(\mathcal{I}_*) = d_A^2(nd_B^2 - 1) \frac{2}{nd_B \min\{d_A, d_B\}} \longrightarrow \Pi_\infty, \\ \Pi_\infty &:= \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} = 2d_A \max\{d_A, d_B\},\end{aligned}$$

as $n \rightarrow \infty$. Thus the product has a finite large- n limit despite the linear growth of the affine dimension and the $1/n$ decay of the centred radius; it is not claimed to be exactly invariant under outcome branching at finite n .

Proof. The limit follows from the displayed explicit formula:

$$\begin{aligned}\Pi_n &= \frac{2d_A^2(nd_B^2 - 1)}{nd_B \min\{d_A, d_B\}} = \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} - \frac{2d_A^2}{nd_B \min\{d_A, d_B\}} \\ &\xrightarrow{n \rightarrow \infty} \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} = 2d_A \max\{d_A, d_B\}.\end{aligned}$$

$\square$

Corollary 5.19 (Scaling with the number of outcomes). *Fix d_A, d_B and consider latent dimensions r_n with $r_n/n \rightarrow c$.*

1. *If $c < d_B^2$, then $r_n \leq nd_B^2 - 2 \leq L_{A,B}^{(n)}$ eventually, so $\beta_{r_n}^{\text{best},(n)} = 1$ for all large n : the certified lower bound saturates at its maximal value 1 already through flagged output preparation (in particular, for every fixed r one has $\delta_{r, \text{inst}}^\diamond \geq 1$ for all large n).*
2. *If $d_B^2 < c < d_A^2 d_B^2$, then eventually $L_{A,B}^{(n)} < r_n < D_{A,B}^{(n)}$ (since $L_{A,B}^{(n)} = nd_B^2 + O(1)$ and $D_{A,B}^{(n)} = nd_A^2 d_B^2 + O(1)$), so the certified lower bound is $\beta_{r_n}^{\text{best},(n)} = \gamma_{A,B}^{(n)} \rightarrow 1/d_A$ — a constant independent of n , strictly better than the naive $1/n$ -decaying ball bound — while every admissible upper bound remains at least 1; the certified bracket therefore has asymptotic length at least $1 - 1/d_A$ in this regime.*
3. *If $c > d_A^2 d_B^2$, then $r_n \geq D_{A,B}^{(n)}$ eventually and $\delta_{r_n, \text{inst}}^\diamond = 0$ exactly.*

Thus the number of outcomes linearly enlarges the parameter count needed for exact reconstruction while dividing the centred positivity margin by n , and the intermediate obstruction carries a constant, n -independent component $1/d_A$.

Remark 5.20 (Error-one barrier, scoped). *For $r < D_{A,B}^{(n)}$ every admissible term of the upper envelope presented in this paper is at least 1 (Proposition 5.13). Consequently the three constructions of this paper — constant codes, linear/convex projection with Helly selection, and balanced block pinching — cannot achieve error $\varepsilon < 1$ in the intermediate regime $L_{A,B}^{(n)} < r < D_{A,B}^{(n)}$. This is a statement about the specific constructions presented here, not a universal impossibility; nonlinear manifold encodings or other families of decoders are not excluded.*

Three candidate tools, however, do not break the barrier, and it is worth recording why. (i) Covering-number encodings. An ε -net of a D -dimensional convex body has size $N(\varepsilon) \geq (r_0/\varepsilon)^D$ up to constants, where r_0 is the radius of any ball inscribed in the body (which exists by full affine dimension, and transfers between equivalent norms), so a net-based code needs $\Theta(D \log_2(1/\varepsilon))$

bits — more than D bits for every $\varepsilon < r_0/2$. The affine encoding already achieves error 0 with exactly D coordinates (Theorem 5.7); nets are therefore strictly dominated by affine coordinates and cannot certify sub-one error below the affine threshold. (ii) Orthogonal joins of separation-two spheres. These are precisely the constructions already absorbed into $\beta_r^{\text{best},(n)}$ in Section 4: replacement-only joins are maximised by the single-block partition (Remark 4.15, giving L_{rep}), and the join mechanism of Proposition 4.12 admits at most one input-dependent sphere (Remark 4.13, giving L_{join}). (iii) Re-centred balls. The intermediate certificate ρ_n is realised by the radial construction about $\mathcal{I}_*$ (Theorem 3.3); by Theorem 3.12 no centre of the instrument body carries a larger relative ball, so re-centring cannot push the ball mechanism past ρ_n . Within mechanisms (i) and (ii) the value 1 is certified up to $L_{A,B}^{(n)}$ and not beyond; whether some other construction extends the separation-two regime past $L_{A,B}^{(n)}$ is open (Remark 2.6(b)), and by (iii) any improvement of the intermediate lower bound $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$ must come from mechanisms other than inscribed balls.

6 Specialisations, examples, and explicit trade-off

Table 1 summarises how the exact instrument parameters of the preceding sections specialise to channels and POVMs.

- **Channel case** $n = 1$. The instrument body reduces to the channel body, and Theorem 3.4 gives, for $d_B > 1$,

$$\rho_{\diamond}(\Omega_{A,B}) = \frac{2}{d_B \min\{d_A, d_B\}},$$

with covering radius $2(1 - 1/(d_B \min\{d_A, d_B\}))$, global in-radius equal to the centred value by Theorem 3.12, and zero-error threshold $d_A^2(d_B^2 - 1)$. (All normalisations are the ones fixed in Remark 2.1; no comparison with external conventions is asserted.)

- **POVM case** $d_B = 1$, $n \geq 2$. Every instrument has the form $\mathcal{I}_k(X) = \text{Tr}(E_k X)$ with $E_k \geq 0$, $\sum_k E_k = I_A$ (Remark 2.4). The centred in-radius is $\rho_{\diamond, \text{inst}}(\mathcal{I}_*) = 2/n$, and this is also the global in-radius over all centres (Theorem 3.12); the flagged replacement sphere has dimension $n - 2$, and the affine dimension is $d_A^2(n - 1)$.
- **Input-independent case** $d_A = 1$. The map

$$\mathcal{I}_k(x) = x \tau_k, \quad x \in \mathbb{C}, \quad \tau \in \mathcal{S}_{\text{flag}}(B \otimes C_n),$$

is an isometry $\text{Inst}_n(\mathbb{C}, B) \cong \mathcal{S}_{\text{flag}}(B \otimes C_n)$ with

$$\|\vec{\mathcal{R}}_{\tau} - \vec{\mathcal{R}}_{\sigma}\|_{\diamond, \text{inst}} = \sum_k \|\tau_k - \sigma_k\|_1,$$

because the input is one-dimensional. Hence all state-space results transfer, and the full-body antipodal profile is exactly

$$\alpha_r(\text{Inst}_n(\mathbb{C}, B)) = \begin{cases} 1, & 0 \leq r \leq nd_B^2 - 2, \\ 0, & r \geq nd_B^2 - 1. \end{cases}$$

The centred in-radius is still $2/(nd_B)$ (since $s = \min\{1, d_B\} = 1$) as a radius of the body inside its affine hull, and by Theorem 3.12 it is also the global in-radius over all centres; it plays no role in the antipodal profile, which is governed by the lower-dimensional replacement structure of the body.

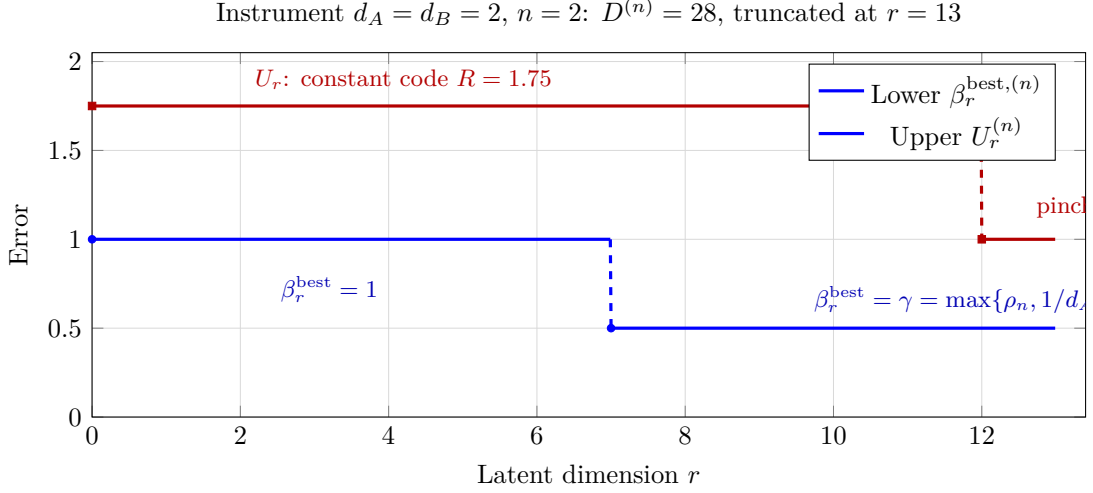


Figure 1: Instrument example of Example 6.1 ($n = 2$, $d_A = d_B = 2$): $nd_B^2 - 2 = 6$, $L_{A,B}^{(n)} = 6$, $D^{(n)} = 28$. Strengthened certified lower envelope: 1 for $0 \leq r \leq 6$, $\gamma = \max\{\rho_n, 1/d_A\} = \max\{1/4, 1/2\} = 1/2$ for $7 \leq r < 28$, 0 at $r \geq 28$. Certified upper envelope on the plotted range: $R_{\text{inst}} = 2(1 - 1/8) = 1.75$ for the constant code on $0 \leq r < 12$, and 1 from balanced two-sector output pinching for $12 \leq r \leq 13$ ($Q_2(2) = 2$, $r_{\text{out}} = 12$). The convex-projection bound exceeds 1.75 on the whole plotted range ($56/29$ at $r = 0$, $34/18$ at $r = 11$), and input pinching enters only at $r_{\text{in}} = 14$, outside the range; the plotted envelope is therefore verified. The plot truncates at $r = 13$ for readability; the full envelope runs to $D = 28$.

- **$1/n$ suppression versus n -fold growth.** The centred in-radius at the uniform centre scales as $1/n$ for fixed d_B , while the affine dimension $D_{A,B}^{(n)} = d_A^2(nd_B^2 - 1)$ grows linearly in n . Thus more outcomes increase the number of parameters needed for exact reconstruction but reduce the local robustness of the uniform instrument; the full-direction bound $\alpha_r \geq 1/d_A$ shows that a part of the intermediate obstruction is independent of n .

Example 6.1 ($d_A = d_B = 2$, $n = 2$). Here $D = 28$, $L_{\text{rep}} = 6$, $L_{\text{obs}} = 2$, $N_{\text{join}} = 2$, $L_{\text{join}} = 3$, so $L_{A,B}^{(n)} = 6$; $\rho_n = 1/4$ and $\gamma_{A,B}^{(n)} = \max\{1/4, 1/2\} = 1/2$. Thus

$$\beta_r^{\text{best},(n)} = \begin{cases} 1, & 0 \leq r \leq 6, \\ 1/2, & 7 \leq r < 28, \\ 0, & r \geq 28. \end{cases}$$

The upper envelope: $R_n = 1.75$; the convex bound equals $56/29 \approx 1.931$ at $r = 0$ and $34/18 \approx 1.889$ at $r = 11$, both above 1.75; balanced two-sector output pinching gives 1 at $r_{\text{out}} = d_A^2(nQ_2(d_B) - 1) = 4(2 \cdot 2 - 1) = 12$ (here $Q_2(2) = 2$); input pinching enters only at $r_{\text{in}} = Q_2(d_A)(nd_B^2 - 1) = 2(8 - 1) = 14$, outside the plotted range. Hence on $0 \leq r \leq 13$ the verified upper envelope is 1.75 for $0 \leq r < 12$ and 1 for $12 \leq r \leq 13$. At $r = 0$ the exact width is 1.75 and the exact gap is $0.75 = 1 - \rho_n$. Figure 1 plots both envelopes on this range, and Table 2 collects further representative parameter sets.

Acknowledgements

The author thanks the anonymous referees for helpful comments.

Quantity	Channel $n = 1$	Instrument $n \geq 1$	Origin
Affine dimension	$d_A^2(d_B^2 - 1)$	$d_A^2(nd_B^2 - 1)$	$nd_A^2d_B^2 - d_A^2$ via surjectivity of the marginal map $(X_1, \dots, X_n) \mapsto \sum_k \text{Tr}_B X_k = H$, e.g. $X_1 = H \otimes I_B/d_B$
Centred in-radius	$2/d_B s$	$2/nd_B s$	$\max\{0, \lambda_{\max}(-X)\} \leq s \ \vec{\Delta}\ _{\diamond, \text{inst}}/2d_A$, $s = \min\{d_A, d_B\}$
Global in-radius	$2/d_B s$	$2/nd_B s$	in-radius concave in the centre (Lemma 3.11); Haar barycentre of $U(B) \times S_n$ orbits is $\mathcal{I}_*$; Jensen (Theorem 3.12)
Covering radius $R(\mathcal{I}_*)$	$2(1 - 1/d_B s)$	$2(1 - 1/nd_B s)$	$\sigma_{EBC_n} \leq nd_B s (\rho_E \otimes I_B/d_B \otimes I_{C_n}/n)$ plus $a \leq Kb$
Replacement sphere	$S^{d_B^2-2}$	$S^{nd_B^2-2}$	$X \mapsto X_+/t_X$, orthogonal Jordan supports
Full-direction sphere	$S^{d_A^2(d_B^2-1)-1}$	$S^{D^{(n)}-1}$	Jordan decomposition plus correction state $\nu = (I_{A_0} - \mu)/(d_A - 1)$
Diamond diameter	2	2	flagged channels have diamond norm 1; orthogonal flagged states attain 2
Zero-error threshold	$d_A^2(d_B^2 - 1)$	$d_A^2(nd_B^2 - 1)$	affine coordinates plus Borsuk–Ulam collision

Table 1: Channel versus instrument exact geometry. The $1/n$ factor is the uniform trace-budget split among n outcomes at the uniform centre; the same numerical positivity budget $2/(nd_B s)$ is realised by two distinct witnesses — an orthogonal-output isometry witness for $d_B > 1$, and an effect-transfer witness $(\pm \mathcal{F}/2)$ for $d_B = 1$, $n \geq 2$ — which live in different faces of the positive cone. Maximality of the radius uses the eigenvalue $1/(nd_A d_B) - t's/(2d_A) < 0$ for $t' > t$, or $1/(nd_A) - t'/(2d_A) < 0$ in the POVM case.

A Finite-program evaluation: background

This appendix records, for completeness and motivation, the no-programming obstruction for unitary channels. It is logically independent of the instrument geometry developed above: the main theorems concern *representational* dimension, whereas this appendix concerns *programmable evaluation* with a fixed processor.

Definition A.1 (Exact finite-program evaluator). *Let E and B be finite-dimensional. An exact finite-program evaluator consists of a finite-dimensional program system M , a completely positive trace-preserving (CPTP) processor $\Gamma : L(M \otimes E) \rightarrow L(B)$, and a program state $\pi_\Phi \in \mathcal{S}(M)$ for every channel $\Phi \in \text{Chan}(E, B)$, such that*

$$\Gamma(\pi_\Phi \otimes \rho) = \Phi(\rho)$$

for every input state ρ on E . By linearity the same identity holds for every input operator.

(d_A, d_B, n)	$D^{(n)}$	L_{rep}	L_{obs}	N_{join}	L_{join}	$L_{A,B}^{(n)}$	$\gamma_{A,B}^{(n)}$
$(2, 2, 2)$	28	6	2	2	3	6	1/2
$(3, 2, 2)$	63	6	7	2	8	8	1/3
$(2, 3, 1)$	32	7	2	1	invalid	7	1/2
$(2, 1, 2)$	4	0	2	0	invalid	2	1

Table 2: Certified one-sphere dimensions and intermediate envelope value, with $L_{A,B}^{(n)} = \min\{D^{(n)} - 1, \max\{L_{\text{rep}}, L_{\text{obs}}, L_{\text{join}}\}\}$ and $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$. For $(2, 1, 2)$ one has $nd_Bs = 2$ and the collapse of Theorem 5.16 applies: $\beta_r^{\text{best}} = 1$ throughout $r < D$.

Theorem A.2 (No finite exact evaluator for unitary channels). *Let $\dim E > 1$. No finite-dimensional program system and fixed CPTP processor $\Gamma : \mathcal{L}(M \otimes E) \rightarrow \mathcal{L}(E)$ can exactly evaluate every unitary channel on E .*

Proof. For pure program states, the no-programming theorem [7] shows that distinct unitary channels require orthogonal programs. It remains to pass from pure to mixed programs. Suppose

$$\pi_U = \sum_j q_j |p_j\rangle\langle p_j|, \quad q_j > 0,$$

is a spectral decomposition of a program that evaluates the unitary channel $\mathcal{U}_U$. For every pure input $|\psi\rangle$, linearity gives

$$U|\psi\rangle\langle\psi|U^\dagger = \sum_j q_j \Gamma(|p_j\rangle\langle p_j| \otimes |\psi\rangle\langle\psi|).$$

Because Γ is CPTP, each summand $\sigma_{j,\psi}$ is a density operator. The right-hand side is a convex decomposition of a pure state, and pure states are extreme points of the state space, so each positive-weight summand equals the same pure output: $\sigma_{j,\psi} = U|\psi\rangle\langle\psi|U^\dagger$ for all j and all pure $|\psi\rangle$. Fixing j , the map $\Lambda_j(X) = \Gamma(|p_j\rangle\langle p_j| \otimes X)$ is a channel agreeing with $\mathcal{U}_U$ on every rank-one projector; rank-one projectors span $\text{Herm}(E)$ over $\mathbb{R}$, and complex linearity extends the equality to all of $\mathcal{L}(E)$. Hence every pure vector in the support of π_U programs $\mathcal{U}_U$ exactly.

Applying the same argument to programs of two distinct unitary channels, the pure-state no-programming theorem makes every positive-weight eigenvector of the first program orthogonal to every positive-weight eigenvector of the second. Those eigenvectors span their respective supports, so $\text{supp } \pi_U \perp \text{supp } \pi_V$. A finite-dimensional program space cannot contain infinitely many mutually orthogonal nonzero supports, while $\dim E > 1$ gives infinitely many distinct unitary channels. This proves the claim. $\square$

Corollary A.3 (Unequal output systems). *Let $\dim E > 1$ and $\dim B \geq \dim E$. There is no finite-dimensional exact evaluator for all channels $E \rightarrow B$.*

Proof. Choose an isometry $V : E \rightarrow B$, let $\iota(\rho) = V\rho V^\dagger$, fix a state ω on E , and define the recovery channel

$$\mathcal{R}(\sigma) = V^\dagger \sigma V + \text{Tr}[(I_B - VV^\dagger)\sigma]\omega,$$

which is CPTP and satisfies $\mathcal{R} \circ \iota = \text{id}_E$. An evaluator for all channels $E \rightarrow B$ would in particular evaluate every channel $\iota \circ \mathcal{U}_U$ for the fixed embedding ι ; postcomposing its processor with $\mathcal{R}$ would then give an exact finite evaluator for all unitary channels on E , contradicting Theorem A.2. $\square$

Corollary A.4 (Vanishing error requires unbounded program dimension). *Fix a system E with $\dim E > 1$. For each j , let M_j be a program system of dimension m_j and let $\Gamma_j : L(M_j \otimes E) \rightarrow L(E)$ be a CPTP processor. Write $\Gamma_{j,\pi}(X) = \Gamma_j(\pi \otimes X)$ and suppose*

$$\sup_{U \in \mathcal{U}(E)} \min_{\pi \in \mathcal{S}(M_j)} \|\Gamma_{j,\pi} - \mathcal{U}_U\|_\diamond \leq \varepsilon_j, \quad \varepsilon_j \rightarrow 0.$$

Then $m_j \rightarrow \infty$.

Proof. Suppose otherwise. Passing to a subsequence, assume $m_j \leq m_{\max}$ for every j . Let $M \simeq \mathbb{C}^{m_{\max}}$ and choose isometries $W_j : M_j \rightarrow M$. Put $P_j = W_j W_j^\dagger$, fix $\tau_j \in \mathcal{S}(M_j)$, and define the CPTP maps

$$\Lambda_j(\eta) = W_j^\dagger P_j \eta P_j W_j + \text{Tr}[(I_M - P_j)\eta] \tau_j,$$

so that $\Lambda_j(W_j \pi W_j^\dagger) = \pi$ for $\pi \in \mathcal{S}(M_j)$. The extended processors $\tilde{\Gamma}_j = \Gamma_j \circ (\Lambda_j \otimes \text{id}_E) : L(M \otimes E) \rightarrow L(E)$ are CPTP and preserve every originally programmed channel. The set of CPTP maps on the fixed spaces $M \otimes E \rightarrow E$ is compact, so, passing to a further subsequence *chosen independently of the target unitary*, $\tilde{\Gamma}_j \rightarrow \Gamma$ in diamond norm. Now fix a unitary U . For each j choose $\pi'_j(U) \in \mathcal{S}(M_j)$ with approximation error at most ε_j (such a minimiser exists by compactness of $\mathcal{S}(M_j)$ and continuity of the induced channel), and embed it as $\pi_j(U) = W_j \pi'_j(U) W_j^\dagger \in \mathcal{S}(M)$. By compactness of $\mathcal{S}(M)$ there is a U -dependent subsubsequence with $\pi_j(U) \rightarrow \pi_U$; along it the processors still converge to the same Γ . Continuity of the evaluation map in both processor and program state gives $\Gamma(\pi_U \otimes \cdot) = \mathcal{U}_U$. Since U was arbitrary, the single limiting processor Γ exactly evaluates every unitary channel, contradicting Theorem A.2. Hence no bounded subsequence of (m_j) exists, and since (m_j) is integer-valued this forces $m_j \rightarrow \infty$. $\square$

Remark A.5 (Scope of the no-programming obstruction). *The result forbids a scheme that is simultaneously exact, deterministic, universal over the unitary promise class, and finite-dimensional. It does not forbid approximate, probabilistic, restricted-family, infinite-dimensional, or unbounded-register schemes, nor schemes with an externally supplied classical description. A finite classical program register is still a finite-dimensional program and is not exempt. Storing a Choi operator as a mathematical state is also distinct from evaluating the represented channel with one fixed CPTP processor. Optimal asymptotic program-cost scaling for approximate universal unitary programming is established in [8]; the compactness corollary above records the qualitative divergence consequence needed for motivation.*

References

- [1] M.-D. Choi, *Completely positive linear maps on complex matrices*, Linear Algebra Appl. **10**, 285–290 (1975).
- [2] A. Jamiołkowski, *Linear transformations which preserve trace and positive semidefiniteness of operators*, Rep. Math. Phys. **3**, 275–278 (1972).
- [3] E. B. Davies and J. T. Lewis, *An operational approach to quantum probability*, Commun. Math. Phys. **17**, 239–260 (1970).
- [4] M. Ozawa, *Quantum measuring processes of continuous observables*, J. Math. Phys. **25**, 79–87 (1984).
- [5] G. Chiribella, G. M. D’Ariano, and P. Perinotti, *Realization schemes for quantum instruments*, J. Math. Phys. **50**, 042101 (2009).
- [6] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018).

- [7] M. A. Nielsen and I. L. Chuang, *Programmable quantum gate arrays*, Phys. Rev. Lett. **79**, 321–324 (1997).
- [8] Y. Yang, R. Renner, and G. Chiribella, *Optimal universal programming of unitary gates*, Phys. Rev. Lett. **125**, 210501 (2020).
- [9] J. Matoušek, *Using the Borsuk–Ulam Theorem*, Springer (2003).
- [10] A. Barvinok, *A Course in Convexity*, Graduate Studies in Mathematics **54**, AMS (2002).